

Selected Topics in Nonlinear System Identification

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*Get ready to dive deep into the world of nonlinear systems theory,
where mathematical creatures called kernels bring life to nonlinear systems
in ways even Dr. Frankenstein would envy!*

Chapter 1

An intuitive introduction to nonlinear systems

Once upon a time, the scientific community created linear systems theory. The ultimate goal of this creation was to have a solid theoretical framework that enables one to understand, describe, predict and control the linear dynamic behaviour of a system. As linear systems theory makes mathematical abstraction of the nature of a system, it is applicable in different domains such as signal processing, automatic control, telecommunications, etc.

Unfortunately, systems in practice are not always linear. Practical implementations are often subjected to nonlinear behaviour up to some level that degrades the performance of the resulting system. For nonlinear systems, it is difficult, and in most cases even impossible, to derive and evaluate an exact mathematical description. Does this mean that you need to throw away all knowledge you gathered up to now about linear systems? Luckily, the answer to this question is no - phew! The reason is that most nonlinear systems can be sufficiently well approximated by a linear model that describes a large portion of the dynamic behaviour. Nevertheless, it is still meaningful for an engineer to intuitively understand how nonlinear phenomena manifest themselves. For that reason, the goal of this course is to offer a good basis to recognize, understand, model, and deal with nonlinear behaviour.

To obtain a solid primer in nonlinear system analysis and modelling, this course incorporates the following topics

- Chapter 1 repeats the basic principles of linear systems theory. Afterwards, an intuitive explanation is provided about how nonlinear behaviour manifests itself and how one can detect the presence of it.
- Chapter 2 introduces the concept and underlying assumptions of linear approximation. Both the [Describing Function \(DF\)](#) and [Best Linear Approximation \(BLA\)](#) theory are discussed in detail.
- Chapter 3 details how one can model the time- and frequency-domain input-output behaviour of a nonlinear system with the Volterra series approximation.
- Chapter 4 concludes with several application examples where nonlinear analysis and modelling is critical to reach the application objectives.

Further reading

Nonlinear system identification is a popular and multi-disciplinary research topic that has a lot of applications. Therefore, a vast amount of literature is available about this topic. If you are interested in the topic and want additional information you can consult reference works listed below. Note that it is not the goal that you read and study all these books.

- *System Identification: A Frequency Domain Approach*
by Rik Pintelon and Johan Schoukens.
- *Nonlinear System Identification: From Classical Approaches to Neural networks and Fuzzy Models*
by Oliver Nelles
- *Nonlinear System Identification: NARMAX Methods in the Time, Frequency, and Spatio-Temporal domains*
by Stephen A. Billings
- *The Volterra and Wiener Theories of Nonlinear Systems*
by Martin Schetzen

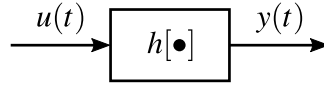


Figure 1.1: General representation of a system $h[\bullet]$. The system is excited by a continuous-time input $u(t)$ that induces a continuous-time response $y(t)$.

1.1 The basic principles

Before diving into the topic of nonlinear systems, we first need to formally introduce the concepts of linearity, nonlinearity, and how they affect the behaviour of a system. The system identification community likes to make abstraction of what a system is such that the developed notions and techniques are generally applicable irrespective of the application area. You can think of a system as a part of the physical world that interacts with its surroundings through a well-defined number of ports. The interaction is described by signals that are either forced on (inputs) or appear at (outputs) these ports. The system here is represented by the mathematical operation $h[\bullet]$ (Figure 1.1). Most systems considered here have a continuous-time (CT) input port that is excited by the signal $u(t)$. The system processes this signal and this results in a corresponding continuous-time output response $y(t)$ at its output port. Formally, we represent the system as

$$y(t) = h[u(t)] \quad (1.1)$$

A linear system is a system that satisfies the superposition and homogeneity principle. These principles state that if the input $u_1(t)$ produces the response $y_1(t)$, and the input $u_2(t)$ produces the response $y_2(t)$; then the linear combination $\alpha u_1(t) + \beta u_2(t)$ will produce $\alpha y_1(t) + \beta y_2(t)$ with $\alpha, \beta \in \mathbb{R}$. We furthermore assume that all the considered systems possess the properties of time-invariance and causality.

Time-invariance expresses that the behaviour of the system does not change over time

$$y(t) = h[u(t)] \quad \rightarrow \quad y(t - \tau) = h[u(t - \tau)] \quad \forall \tau \in \mathbb{R} \quad (1.2)$$

Causality expresses that the response of the system taken on time instant t_0 only depends on the past and present values $t \leq t_0$ of its input.

When a system's output depends only on the current value of the input signal, the system is called *static*. When the system is not static it is called *dynamic* and it is said to have memory, because it "remembers" past values of the input and output signal.

The behaviour of a causal linear time-invariant dynamic system is completely described by its impulse response function $g(t)$. The impulse response function relates the output signal $y(t)$ with the input signal $u(t)$ by means of the convolution integral (* is the convolution operator)

$$y(t) = (g * u)(t) = \int_{-\infty}^t g(t - \tau) u(\tau) d\tau \quad (1.3)$$

Equivalently, the system can be described in the frequency domain with the so-called frequency response function $G(j\omega)$ that is obtained by applying the Fourier transform $\mathcal{F}\{\bullet\}$ to $g(t)$

$$G(j\omega) = \mathcal{F}\{g(t)\} = \int_0^{+\infty} g(t) e^{-j\omega t} dt \quad (1.4)$$

The frequency response function relates the frequency-domain output $Y(\omega) = \mathcal{F}\{y(t)\}$ with the input $U(\omega) = \mathcal{F}\{u(t)\}$ by means of a multiplication

$$Y(\omega) = G(j\omega) U(\omega) \quad (1.5)$$

Intuitively, the frequency response describes the magnitude and phase change that a single-tone sinusoid with frequency ω would undergo when it passes through the system

$$u(t) = A \sin(\omega t) \rightarrow y(t) = A |G(j\omega)| \sin(\omega t + \angle G(j\omega)) \quad (1.6)$$

where $|\bullet|$ and $\angle\bullet$ are, respectively, the magnitude and phase operator. This input-output relation is independent of the amplitude A .

Every system that does not satisfy the superposition principle is said to be nonlinear. For nonlinear systems, it is not possible to generalise the response to a specific class of input signals to another class of input signals. The behaviour of a nonlinear system therefore becomes signal dependent. As a result, it is no longer feasible to use one single metric such as the impulse response or transfer function to fully describe the behaviour of the system. This makes the analysis and identification of nonlinear systems much more complicated than that of linear systems. You will discover in Chapter 3 how to generalize the notion

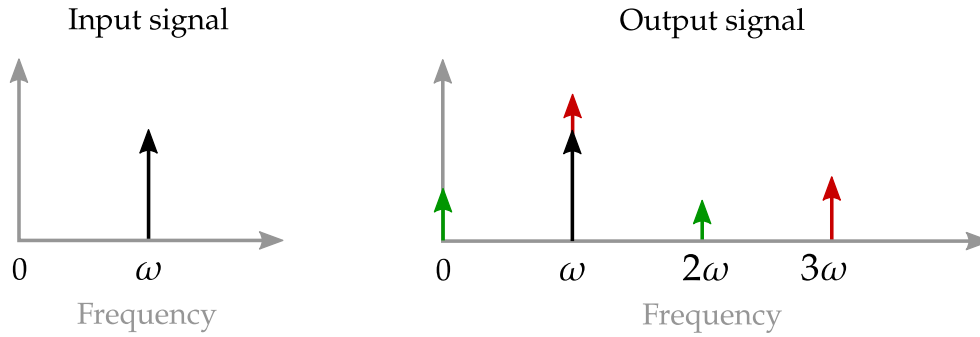


Figure 1.2: Application of a complex exponential signal to a nonlinear system creates harmonics at the output.

of impulse response and transfer function for nonlinear systems. But before we start with this mathematical endeavour, the remainder of this chapter demonstrates how nonlinear behaviour manifests itself and how one can detect it.

1.2 Classical methods to verify the (non)linearity of a system

From your basic course on Measurement and Identification you learned that steady-state sinusoidal (or more general complex exponential) excitations are preferred if you want to characterize the linear dynamic behaviour of a system. The reason for this is that complex exponentials are eigenfunctions of a linear time-invariant system. Application of such a complex exponential only introduces an amplitude and phase change, as illustrated in (1.6). Therefore, the input and output of a linear time-invariant system have the *same* spectral components in steady state.

But what would happen if we applied such a complex exponential signal to a nonlinear system? Let us consider as an example an elementary static nonlinearity $y = u + u^2 + u^3$ that contains quadratic and cubic nonlinear effects to formulate an answer to this question. Application of a steady-state input excitation $u(t) = 2 \cos(\omega t) = e^{j\omega t} + e^{-j\omega t}$ results in the following output contributions for the quadratic nonlinearity

$$\begin{aligned} y_{\text{quadratic}}(t) &= (e^{j\omega t} + e^{-j\omega t})^2 \\ &= e^{j2\omega t} + e^{-j2\omega t} + 2 \end{aligned} \tag{1.7}$$

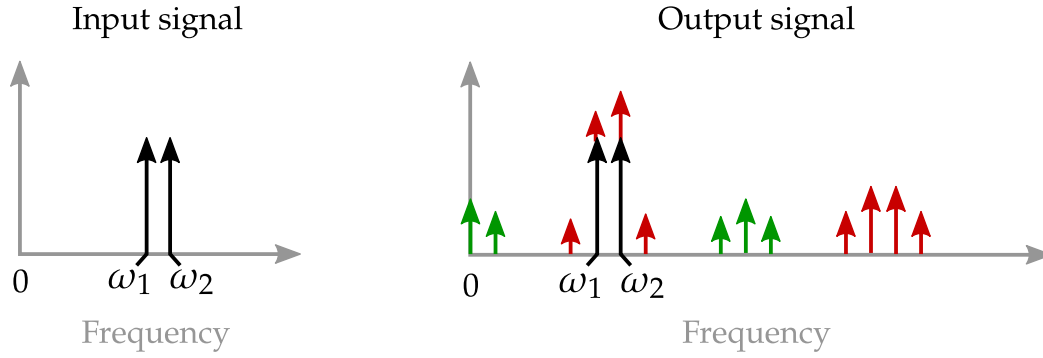


Figure 1.3: With a two-tone excitation, intermodulation contributions are created that are close to the original tones.

and for the cubic nonlinearity

$$\begin{aligned} y_{\text{cubic}}(t) &= (e^{j\omega t} + e^{-j\omega t})^3 \\ &= e^{j3\omega t} + 3e^{j\omega t} + 3e^{-j\omega t} + e^{-j3\omega t} \end{aligned} \quad (1.8)$$

We can directly observe that these nonlinearities do not preserve the spectral content of the excitation signal as a linear system would do (Figure 1.2). At the output of the quadratic nonlinearity, spectral energy appears in the output spectrum at -2ω , 2ω and DC. The cubic nonlinearity on the other hand generates harmonic distortion at -3ω and 3ω but also retains energy at the original excited frequency. All these spectral components at harmonics of the original frequency ω are not present in the input excitation. This phenomenon of frequency recombination is a general characteristic of any nonlinear system that preserves the periodicity of the input excitation at the output. Therefore, evaluation of the presence of harmonics is a first adequate test to verify the (non)linearity of a system.

Using the harmonics to determine the nonlinear behaviour can potentially result in erroneous conclusions. This is most often the case for radiofrequency (RF) systems where the harmonics are filtered out due to dynamic effects present in the system. If extensive filtering is present, it is common practice to apply a two-tone sinusoidal excitation to quantify the in-band nonlinear distortion

$$u(t) = \frac{A}{2} \sin(\omega_1 t) + \frac{A}{2} \sin(\omega_2 t) \quad (1.9)$$

where the two frequencies ω_1 and ω_2 are usually placed close together. When such a two-tone is applied to a nonlinear system, both harmonics and intermodulation products

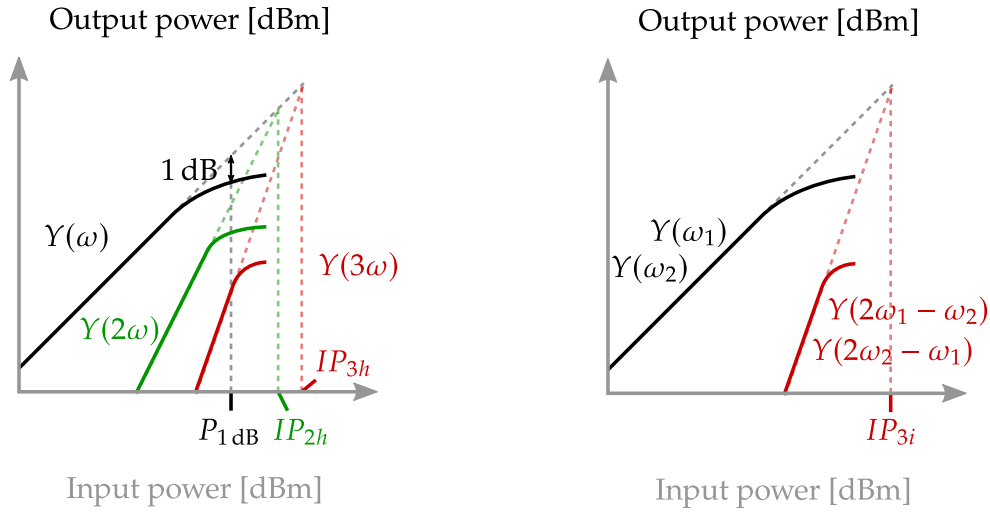


Figure 1.4: Plotting the power of the different tones of the output signal versus the input power, both on a logarithmic scale, reveals the effects of nonlinear behaviour. On the left plot, the result for a single-tone applied to a weakly non-linear circuit is shown. On the right, a two-tone excitation is used. From these plots, measures for linearity, such as the compression and intercept points can be obtained.

appear at the output. The cubic nonlinearity creates tones at $2\omega_1 - \omega_2$ and $2\omega_2 - \omega_1$, which are close to the original set of tones. As a first approximation, they undergo the same filtering as the original two-tone excitation and can be used to get a reliable estimate of the nonlinear behaviour.

An additional test to quantify the nonlinear behaviour of a system is performed by sweeping the input power while the excitation frequency is kept constant. Valuable information about the system is obtained by visualising the power of the tones in the output signal on a log-log plot (Figure 1.4). In a weakly non-linear system, the fundamental has a linear slope of 1 : 1 for small amplitudes. The second and third harmonic have 2 : 1 and 3 : 1 slopes on the log-log scale respectively. From the swept input power data, the 1 dB compression point ($P_{1\text{dB}}$) and intercept points (IP_{2h} , IP_{3h} , and IP_{3i}) are obtained.

1.3 Prediction of the frequency locations in the output spectrum

As became apparent in the previous section, application of a two-tone signal to a nonlinear system creates additional tones at well-specified locations in the output spectrum. These locations are not arbitrarily chosen by the nonlinearity but can be predicted in advance. If this would not be the case, nonlinear testing in general would be impossible as we would not know beforehand which frequencies to expect. It is therefore a useful skill that you can quickly predict these frequency locations.

One of the fundamental expressions of nonlinear behaviour is frequency recombination. Chapter 3 will provide in detail the mathematical basis for this recombination mechanism. However, if you know the input excitation tones and the order of the nonlinearity, you can intuitively predict the recombination outcome. From linear system theory, we know that a convolution performed in the time domain becomes a multiplication in the frequency domain and vice versa. A static polynomial nonlinearity ($y = u^k$) does nothing else than multiplying the input signal several times with itself in the time domain. These multiplications translate into convolutions of the spectral lines in the frequency domain. Such a frequency-domain convolution sums the frequencies of the input tones to create output contributions at different frequencies. The higher the nonlinearity order, the more convolutions are required, and the more frequency combinations appear at the output. In general, a k^{th} order nonlinearity sums the frequencies of k input spectral lines. The output of the nonlinear system thus contains tones at all possible combinations of input frequencies taken k at a time with repetition.

To make the explanation above more concrete, let us reconsider the quadratic and cubic nonlinearities from the previous section that are excited by a real single-tone excitation. Don't forget that a real sinusoidal excitation with a frequency ω always contains two contributions in its spectrum: one at $+\omega$ and one at $-\omega$. For the quadratic nonlinearity, two frequencies at a time need to be selected from $(-\omega, \omega)$ and summed together. In total, $2^2 = 4$ combinations are possible. The cubic nonlinearity on the other hand selects three frequencies with repetition from $(-\omega, \omega)$ and sums these frequencies together. This results in a total of $2^3 = 8$ possible combinations. Remark that the results of the summation are not unique. Often, the recombination process results in the same output frequency. Contributions with the same output frequency are summed together to create one larger

Possible frequency combinations

$$y = u^2$$

$$+\omega + +\omega = +2\omega$$

$$+\omega + -\omega = 0$$

$$-\omega + +\omega = 0$$

$$-\omega + -\omega = -2\omega$$

$$y = u^3$$

$$+\omega + +\omega + +\omega = +3\omega$$

$$+\omega + +\omega + -\omega = +\omega$$

$$+\omega + -\omega + +\omega = +\omega$$

$$+\omega + -\omega + -\omega = -\omega$$

$$-\omega + +\omega + +\omega = +\omega$$

$$-\omega + +\omega + -\omega = -\omega$$

$$-\omega + -\omega + +\omega = -\omega$$

$$-\omega + -\omega + -\omega = -3\omega$$

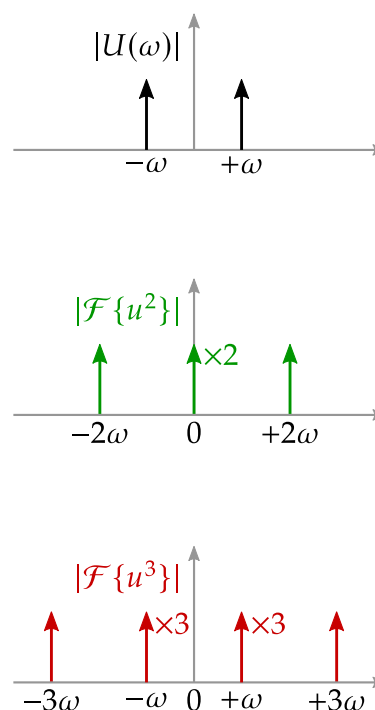


Figure 1.5: A nonlinear system recombines the input tones to create output contributions at different output frequencies.

contribution in the eventual output spectrum as evidenced by (1.7) and (1.8). By means of exercise, can you predict the possible frequency combinations for a quadratic nonlinearity that is excited by the two-tone excitation from Section 1.2?

1.4 Case study of a nonlinear power amplifier

Within this course several theoretical concepts and identification techniques of nonlinear systems are introduced and explained. By means of illustration, we will provide an in-depth, detailed examination of a nonlinear power amplifier. The reason we chose to study such a system is that the nonlinear analysis and modelling of power amplifiers is still a very actual topic in telecommunications, especially with the current transition to mm-wave frequencies in 5G and post-5G technologies.

The power amplifier under study is a ZFL-11AD+ amplifier from Mini-Circuits. It operates

between 2 MHz and 2 GHz and has a typical output 1 dB-compression point of -3.5 dBm. In Section 1.2, you learned that classical methods to evaluate the nonlinear behaviour of such an amplifier use simple excitation signals like one- or two-tones. However, most real-world nonlinear systems are not excited by such elementary signals. This amplifier, for instance, is specifically constructed for cellular, satellite, and GPS communication. It therefore processes modulated signals that carry information. These signals occupy a certain bandwidth around a carrier frequency ω_c

$$u(t) = A(t) \sin(\omega_c t + \varphi(t)) \quad (1.10)$$

The information is encapsulated in the time variation of the amplitude and phase of the signal. The modulation and coding part of the communication standard strives to make $A(t)$ and $\varphi(t)$ as random as possible to maximize the information entropy. Therefore, the content of the data that is being sent is not exactly known in advance and both $A(t)$ and $\varphi(t)$ are random signals.

Due to the modulation with random signals, the exact excitation signal $u(t)$ cannot be known. However, the statistical information, such as the [Power Spectral Density \(PSD\)](#) and [Probability Density Function \(PDF\)](#) are known and specified in the communication standard. A wireless Local Area Network (LAN) signal, for example, uses Orthogonal Frequency Domain Multiplexing (OFDM) modulated signals that occupy a bandwidth of 20 MHz around the carrier frequency of 2.4 GHz. Due to the properties of OFDM modulation, the PDF of the signal resembles a Gaussian. The situation is similar in low-frequency electronic circuits: instead of one- two-tones, the circuits usually have to process noise-like signals in their real-world operation. We cannot predict in advance the exact signals that pass through the circuit, but we can get a good idea of their [PSD](#) and [PDF](#). We also consider these noise-like low-frequency signals as modulated signals, even though they are not represented by a modulated carrier frequency.

Let us now, as an example, take a look at how the power amplifier behaves in response to such a modulated signal. To be able to do so, we need to use a measurement setup that can both apply and acquire IQ-modulated signals. One device that provides this measurement capability is the so-called [Vector Signal Transceiver \(VST\)](#) (Figure 1.6). A [VST](#) consists of two major subcomponents: a transmitter and a receiver. The transmitter takes as inputs an in-phase (I) and quadrature (Q) digital bitstream. Each bitstream is first converted to an analog signal by a synchronized [Digital-to-Analog Converter \(DAC\)](#) that operates at a sampling frequency f_s . Next, the analog signals are filtered and applied to

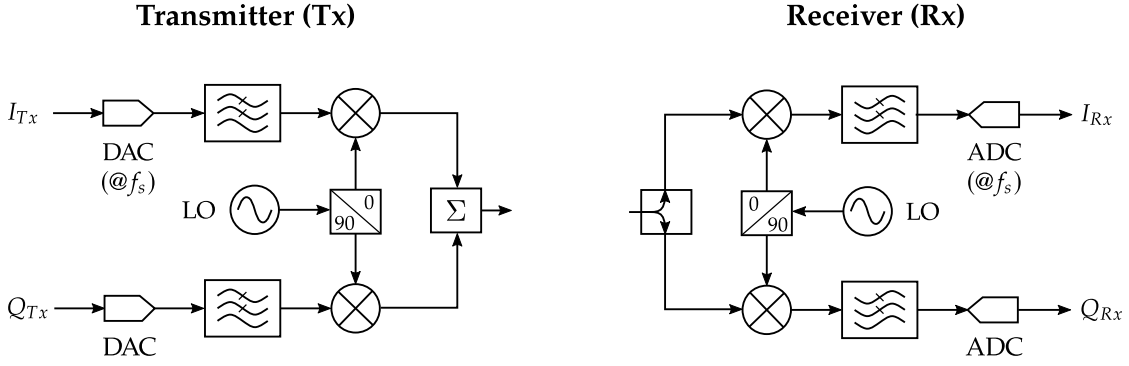


Figure 1.6: A VST consists of a transmitter (Tx) and receiver (Rx) that perform the **Radio-Frequency (RF)** up- and down-conversion, analog-to-digital and digital-to-analog conversion, and lowpass filtering.

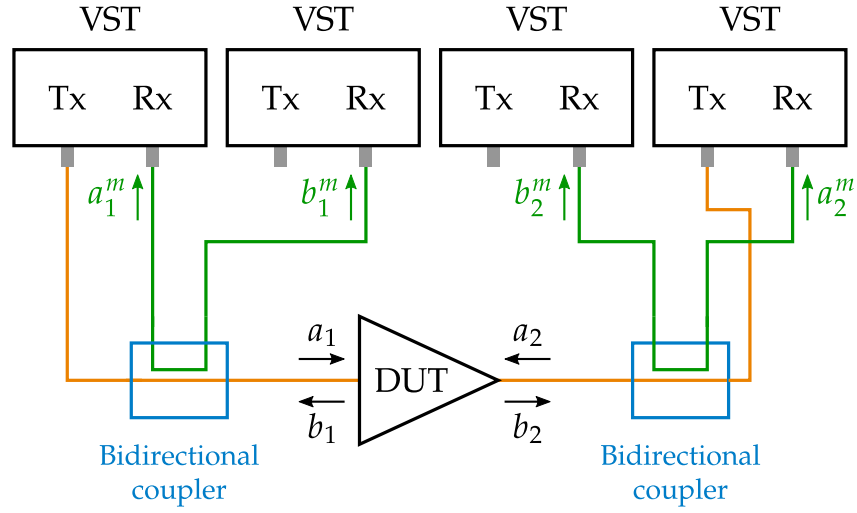
an IQ-modulator, which upconverts the low-frequency modulation content to the **Local Oscillator (LO)** frequency. The receiver performs the opposite operation and uses an IQ-demodulator combined with two synchronized **Analog-to-Digital Converters (ADCs)** to acquire the measurement data. Combining several **VSTs** into one measurement setup enables to characterize the behaviour of the power amplifier under modulated excitation (Figure 1.7).

The goal of the **VST**-based measurement setup is to characterize the scattering parameters (S-parameters) as a function of the power of the modulated excitation signal. The S-parameters are a widely used description in the microwave community and describe the linear relationship between the incoming waves (a_i) and outgoing waves (b_i). As the power amplifier is a two-port device, the general S-parameter formalism boils down to the following mathematical description for our device under test in the frequency domain

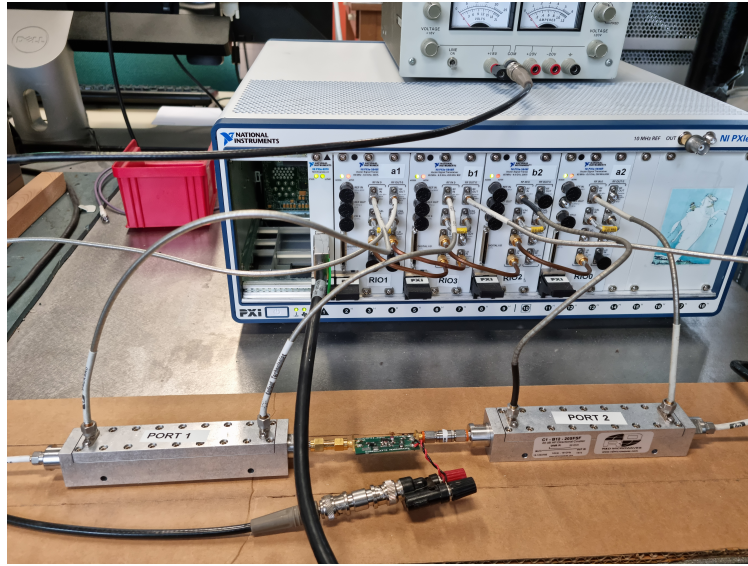
$$\begin{bmatrix} b_1(\omega) \\ b_2(\omega) \end{bmatrix} = \begin{bmatrix} S_{11}(j\omega) & S_{12}(j\omega) \\ S_{21}(j\omega) & S_{22}(j\omega) \end{bmatrix} \begin{bmatrix} a_1(\omega) \\ a_2(\omega) \end{bmatrix} \quad (1.11)$$

To be able to measure the incident and reflected waves, bidirectional couplers are employed that sample the waves at each port. In total four **VSTs** are needed to sample all the waves simultaneously. All the sampling clocks of the different **VSTs** are synchronized and the **LO** is shared to obtain leakage-free measurements.

A measurement campaign has been set up where the input power of the excitation is swept from -20 dBm to 5 dBm in steps of 1 dB. The excitation signal has a 3 MHz modu-



(a) Measurement schematic



(b) Measurement setup

Figure 1.7: A *VST*-based measurement setup is used to measure the incident (a_i) and reflected waves (b_i) at each port of the power amplifier.

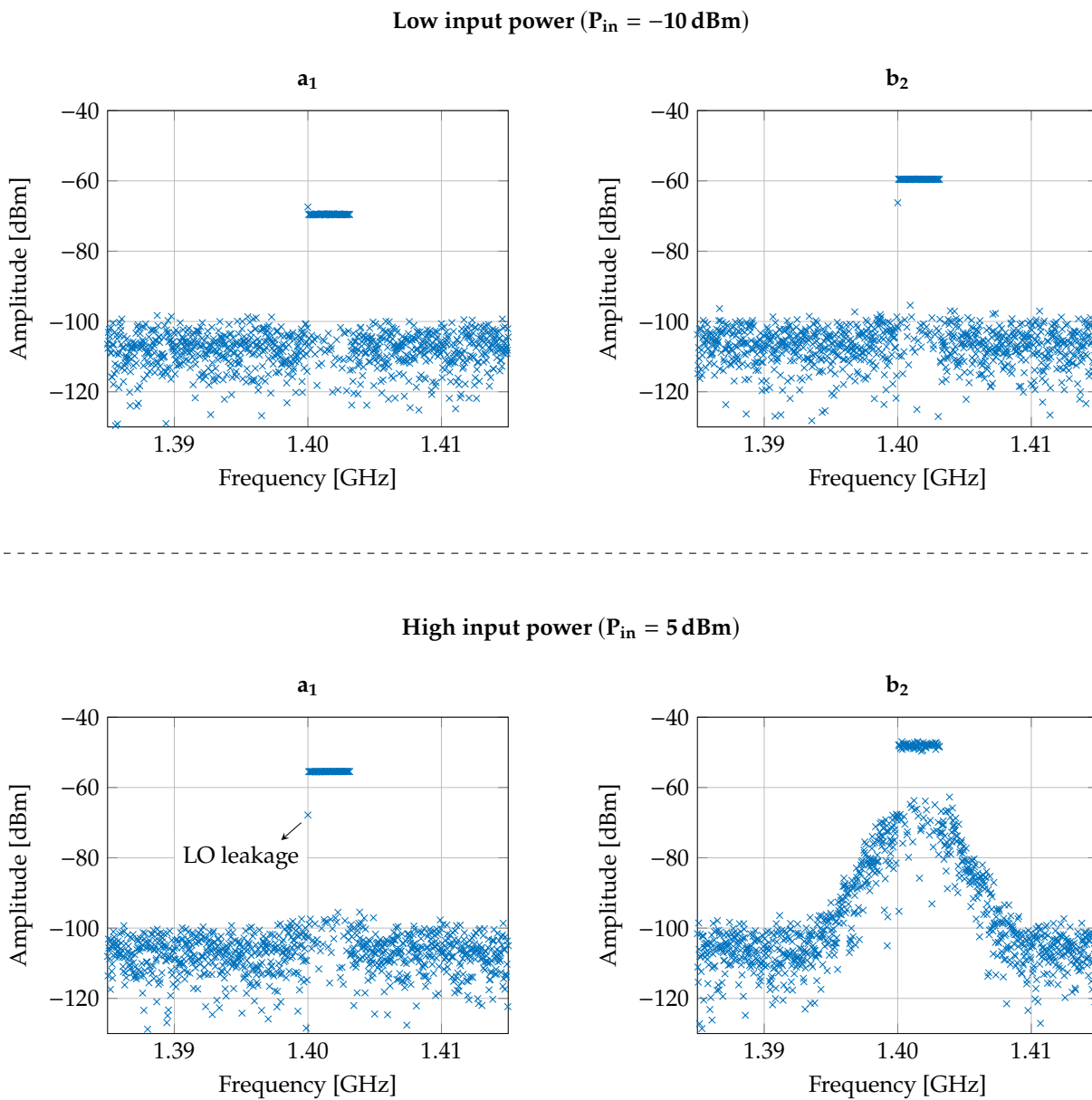


Figure 1.8: At low input power the power amplifier behaves linearly and exhibits no nonlinear effects. At high input power, compression takes place that generates spectral regrowth at frequencies in the vicinity of the excitation bandwidth.

lation bandwidth centered around a 1.4 GHz carrier frequency. Once the measurements are performed, it is important to always perform a sanity check before doing any data processing. Figure 1.8 shows the amplitude of a_1 and b_2 as a function of frequency for two different power level settings (−10 dBm and 5 dBm). Inspection of the raw wave data shows that leakage-free measurements are obtained except for the LO frequency. The reason for this is that the sharing of the LO among the different VSTs inevitably introduces a phase difference, which in turn results in an inaccurate measurement result at the carrier frequency only. The measurement noise floor is situated at around −110 dBm. For the low input power level, the power amplifier behaves as expected. It linearly amplifies the power at the excited frequencies with an amplification factor of approximately 10 dB. The b_2 measurement data for the high input power level shows significant spectral regrowth outside the excitation bandwidth caused by nonlinear compression. Furthermore, the response of the amplifier at the excited frequencies appears to be ‘noisy’. This ‘noise’ does not originate from the measurement uncertainty but originates from the amplifier’s nonlinear behaviour that degrades the signal quality.

Given the acquired measurement data, the question arises how to use it to describe the behaviour of the power amplifier. For low input power levels it is clear that linear S-parameters can still be used as the nonlinear distortion levels are buried under the measurement noise floor. For higher input power levels, this is no longer the case and we shall have to resort to nonlinear characterization and modelling techniques. The pursuit for these techniques is the subject of the succeeding chapters. In each chapter you will discover how you can more accurately describe the nonlinear behaviour of this power amplifier.

Chapter 2

The best linear approximation theory for modulated signals

In the previous chapter, you have learned that classical methods for nonlinear verification use simple excitation signals like one- or two-tones. Moreover, measurements conducted with modulated signals on a power amplifier have revealed that nonlinear behaviour significantly degrades the signal quality. Generally speaking, you require a nonlinear model to completely describe the nonlinear behaviour of the device under test. Luckily it is still possible to acquire information about the linear(ized) behaviour of the device by means of linear approximations. Linear approximations have been introduced to make the nonlinear complexity tractable. They re-enable the use of linear system theory while incorporating signal dependency. These approximations are not unique. In this chapter, we discuss the [BLA](#) theory that has been developed with the purpose of describing the linearized behaviour of nonlinear systems that are excited by modulated signals. The discussion involves the basic principles, underlying assumptions and application examples of the [BLA](#) theory.

2.1 Small- or large-signal linearisation

A possible approach to study nonlinear systems uses simplified models that substitute each nonlinear operation by an approximating linear one. Linear system theory can then

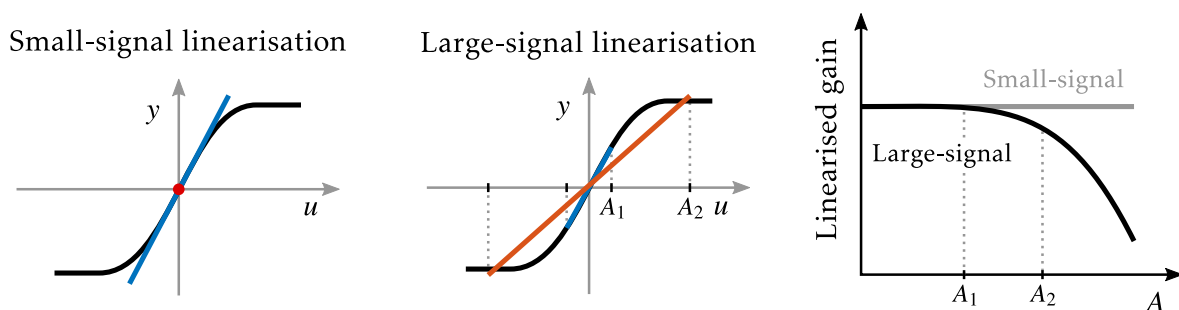


Figure 2.1: Small- (left) and large-signal (middle) linearisation result in a different behaviour as a function of the input signal's properties (right).

be used to predict and analyse the behaviour of the approximating system. As one could expect, the approximation error between these linear results and the behaviour of the original nonlinear system depends strongly on the validity of each linear operation.

Linear approximation can either be performed locally or globally. This boils down to small- and large-signal linearisation respectively. Suppose that we have a static saturating function (Figure 2.1). Small-signal linearisation linearises the function around a nominal operating point, in the case of the left figure this is set to be the origin. Clearly, this linearisation is only valid in the proximity of the operating point. If we would apply a signal u with an input range exceeding the acceptable limits for small-signal operation, the validity of this kind of linear approximation is completely undermined. To cope with this limitation the concept of large-signal least-squares linearisation was introduced. This type of linearisation depends not only on the nonlinear function but also on the properties of the input signal. Different linear approximations are obtained when driven by signals with a different waveform, or even when driven by the same waveform but with a different input range. To demonstrate the usefulness of large-signal linearisation, we consider that we have a single-tone sinusoidal signal with two different amplitudes A_1 and A_2 (Figure 2.1) for which a different approximating function is obtained (— and —). In contrast to the small-signal case where the linearised gain (the slope of the approximating function) is independent of the applied amplitude, the large-signal linearisation does now result in an amplitude-dependent linearised gain and therefore enjoys a substantial advantage over small-signal linearisation at the cost of a larger effort to extract the model. Namely there is no limit to the range of signal amplitudes that can be accommodated. Furthermore, signal-dependent behaviour is introduced as would be the case for a full-blown

nonlinear model. The main difficulty that is encountered with large-signal linearisation is that the resulting approximation depends on both the system and the signal properties (waveform, magnitude, power, spectrum, probability density function, etc.). In practice, assumptions have to be made about the current signal properties to be able to calculate the approximating function. This is where the Best Linear Approximation comes into the picture.

2.2 The Best Linear Approximation

The Best Linear Approximation uses a linear time-invariant dynamic model to linearise the behaviour of the nonlinear system. It therefore considers dynamic nonlinear systems that are excited by stationary random input signals. All signals that possess the same [PSD](#) and the same [PDF](#) result in the same [BLA](#).

The [BLA](#) theory is not applicable to any nonlinear system. The system should have fading memory, which means that it can be approximated arbitrarily well in least-squares sense by a Volterra series (see Chapter 3). An important property of the systems belonging to this system class is that the periodicity of the input signal is preserved. Therefore, systems belonging to this nonlinear class are also called Periodic-In Same Period-Out (PISPO) systems. The periodicity preservation rules out systems that generate sub-harmonics, contain bifurcations, or exhibit chaotic behaviour. It does, however, include systems that show hard nonlinear behaviour such as clipping, dead zones, quantisation, etc.

Consider the the input signal of the nonlinear system is a random signal generated by a stationary process. The input signal is indicated by $u(t)$ and the output signal by $y(t)$. Both signals are defined around their average operating point, such that their mean value equals zero

$$\mathbb{E}\{u(t)\} = 0 \quad \mathbb{E}\{y(t)\} = 0 \quad (2.1)$$

where $\mathbb{E}\{\bullet\}$ is the expected value operator. The [BLA](#) of a nonlinear system is the [Linear Time-Invariant \(LTI\)](#) model g^{BLA} that minimises the mean-square error between the input $u(t)$ and the output $y(t)$. When the [BLA](#) is a stable [LTI](#) system, the mean-square

minimisation can be expressed in the time domain as

$$g^{\text{BLA}} = \underset{g \in \mathcal{G}}{\operatorname{argmin}} \mathbb{E}_u \left\{ |y(t) - g(t) * u(t)|^2 \right\} \quad (2.2)$$

$\mathbb{E}_u \{\bullet\}$ is the expected value operator that is taken with respect to the random variables present within $u(t)$. $\mathcal{G}$ represents the class of all stable LTI models. In general, the BLA should not be a stable linear system. This is why the estimation is best performed in the frequency domain, where it boils down to solving the following minimisation problem

$$G^{\text{BLA}}(j\omega) = \underset{G(j\omega)}{\operatorname{argmin}} \mathbb{E}_U \left\{ |Y(\omega) - G(j\omega)U(\omega)|^2 \right\} \quad (2.3)$$

Equation (2.3) can be solved explicitly and results in the following expression for the BLA

$$G^{\text{BLA}}(j\omega) = \frac{S_{yu}(j\omega)}{S_{uu}(j\omega)} = \frac{\mathbb{E}\{Y(\omega)U^*(\omega)\}}{\mathbb{E}\{|U(\omega)|^2\}} \quad (2.4)$$

$\bullet^*$ is the complex conjugate transpose operator. $S_{yu}(j\omega)$ is the cross-power spectrum between the input and output signals and $S_{uu}(j\omega)$ is the auto-power spectrum of the input signal. They are defined as

$$\begin{aligned} S_{yu}(j\omega) &= \mathcal{F}\{R_{yu}(\tau)\}(j\omega) & R_{yu}(\tau) &= \mathbb{E}\{y(t)u(t-\tau)\} = \int_{-\infty}^{+\infty} y(t)u(t-\tau)dt \\ S_{uu}(j\omega) &= \mathcal{F}\{R_{uu}(\tau)\}(j\omega) & R_{uu}(\tau) &= \mathbb{E}\{u(t)u(t-\tau)\} = \int_{-\infty}^{+\infty} u(t)u(t-\tau)dt \end{aligned} \quad (2.5)$$

Evaluating (2.4) analytically proves to be impossible for dynamic (nonlinear) systems. Therefore, the BLA is predominantly determined non-parametrically at each frequency ω independently from simulated or measured input-output spectra. In many simulation or measurement scenarios, one only has access to time-domain data. These spectra can then be obtained by performing the Discrete Fourier Transform (DFT) on the time-domain signals. Suppose that we have a continuous-time signal $x(t)$ that is sampled equidistantly with a sampling period T_s for a total of N samples at a sampling rate above the Nyquist rate. The DFT $X(k)$ of the sequence $x(nT_s)$ is then defined as

$$X(k) = \text{DFT}\{x(t)\} = \frac{1}{N} \sum_{n=0}^{N-1} x(nT_s) e^{-j\omega_k nT_s} \quad (2.6)$$

In the remainder of this course, we assume that whenever the notation $X(k)$ is used, we are dealing with the **DFT** of the signal $x(t)$ as defined in (2.6). The frequency ω_k that corresponds with the bin k is defined as $\omega_k = 2\pi/(NT_s)$. Remark that the **DFT** requires the time samples to be equally spaced in time. Variable time steps can be used, but require the data to be interpolated to an equidistant time grid before computing the **DFT**. This will introduce interpolation errors and should therefore be handled with care. Using the **DFT** spectra of the input $U(k)$ and output $Y(k)$, the original **BLA** definition in (2.3) is recast to

$$G^{\text{BLA}}(j\omega_k) = \frac{\mathbb{E}_U\{Y(k) U^*(k)\}}{\mathbb{E}_U\{|U(k)|^2\}} \quad (2.7)$$

Apart from estimating $g^{\text{BLA}}(j\omega_k)$, the **BLA** theory additionally provides the means to characterise the contributions that are not related to the linear behaviour of the system. These contributions can be divided into two sources: an additive noise source $n(t)$ and an additive nonlinear distortion source $d(t)$ (Figure 2.2). When the **BLA** is a stable system and no noise is considered, the nonlinear distortion introduced by the nonlinear system can be defined as:

$$d(t) = y(t) - g^{\text{BLA}}(t) * u(t) \quad (2.8)$$

The nonlinear distortion $d(t)$ has the following properties:

- $d(t)$ has zero mean: $\mathbb{E}\{d(t)\} = 0$
- $d(t)$ is uncorrelated with the input signal $u(t)$: $\mathbb{E}\{d(t) u(t - \tau)\} = 0$
- $d(t)$ depends on the input power spectrum
- The autopower spectrum of the distortion $S_{dd}(j\omega)$ is a continuous function of the frequency

These properties indicate that the nonlinear distortion $d(t)$ acts as a noise-like signal. This key observation is used in the next section to estimate the variance corresponding to this contribution.

Example

To showcase some of the fundamental properties of the **BLA**, we analyse a simple static nonlinear system with the following input-output relationship as an example

The nonlinearity is chosen such that it resembles the relationship between the gate voltage

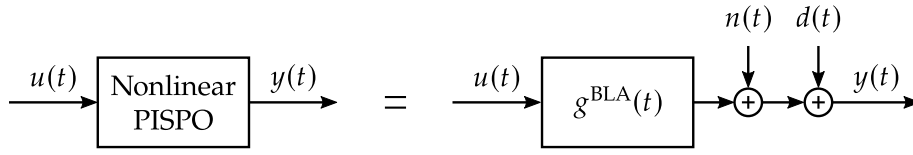
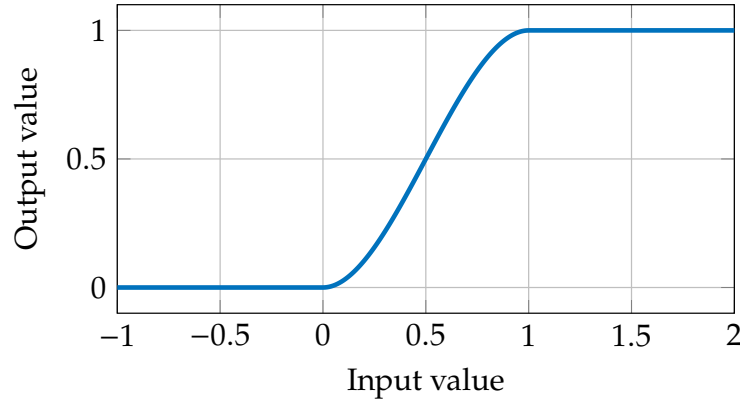


Figure 2.2: Every nonlinear PISPO system that is excited by a random signal can be linearly approximated by the BLA. The unmodelled dynamics are attributed to the noise $n(t)$ and the nonlinear distortion $d(t)$.



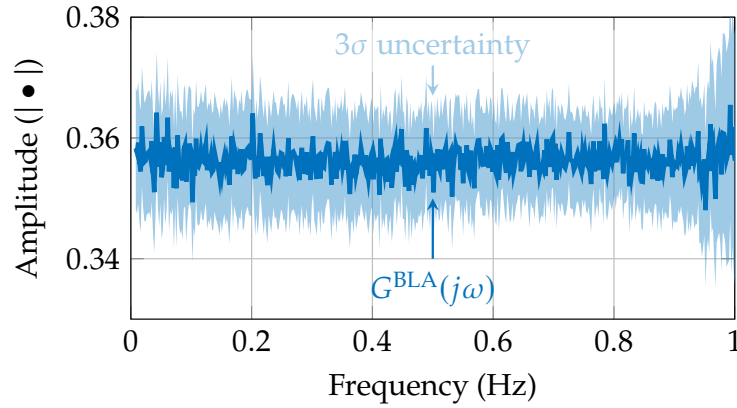
and drain current of a transistor. It is a continuous piecewise function defined as

$$y = g(u) \text{ where } \begin{cases} y = 0 & u < 0 \\ y = 3u^2 - 2u^3 & u \in [0, 1] \\ y = 1 & u > 1 \end{cases} \quad (2.9)$$

The input signal $u(t)$ is a filtered Gaussian independent and identically distributed noise signal around a DC offset of 0.05. Due to the small DC offset, the system behaves as an amplifier biased on the edge of class-B. the [Root Mean Square \(RMS\)](#) value of the signal is 0.1 and most of the power of the noise signal lies in the frequency band between 0 Hz and 1 Hz. Note that the frequency of the excitation signal doesn't matter for a static system like the one we are analysing here.

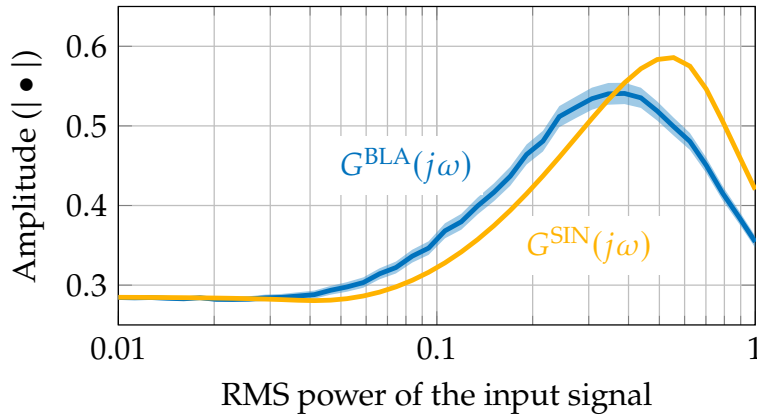
The [BLA](#) is calculated using (2.7) and the estimated value together with its uncertainty are shown below

It is clear that the [BLA](#) is static for this circuit. This is due to the Gaussian distribution of the input signal. If the input signal were to have another [PDF](#), the [BLA](#) for this static



system would not be static. The increase of uncertainty at the end of the frequency band is due to a lower level of the input signal in that frequency region.

As a final test on this example, we show how the BLA changes as a function of the RMS of the input signal. The above procedure is repeated for different RMS values of the input signal. For each RMS value, the magnitude of the BLA at 0.5 Hz and its uncertainty are plotted.



It is clear that the BLA varies strongly as a function of the RMS of the input signal. As a form of comparison, we applied a single-tone excitation with the same RMS and DC offset to the test system

$$u(t) = \sqrt{2} \cdot \text{RMS} \cdot \sin(\omega t) + 0.05 \quad (2.10)$$

We then calculate the resulting output signal $y(t)$ and the large-signal gain of the system as

$$G^{\text{SIN}} = \frac{Y(j\omega)}{U(j\omega)} \quad (2.11)$$

This large-signal gain is then plotted as a function of the amplitude on the same figure of the [BLA](#) above. It is clear that the large-signal linearisation depends on the [PDF](#) of the applied input signals.

2.3 Multisine excitations

Using random, non-deterministic signals as input $u(t)$ for the non-linear system is not always convenient from a practical point of view. In measurement setups and simulations, it is preferred to work with periodic excitation signals. The main reason for this lies in the fact that we can determine the steady-state response of the circuit, which makes it easier to determine the [BLA](#) in the frequency domain. Random signals also complicate the analysis as they introduce spectral leakage in the frequency domain. Furthermore, applying stochastic noise signals to excite the nonlinear PISPO system does not allow us to separate the contributions of $n(t)$ and $d(t)$. Establishing separability of both contributions requires to excite the system with a deterministic signal. The distortion term inherits (non-)determinism from the input signal while the noise term remains stochastic.

The idea is to synthesise a deterministic signal that can mimic the behaviour of the original stochastic signal. As mentioned earlier, the [BLA](#) of a system is equal for two random excitation signals with the same [PSD](#) and [PDF](#). So, to determine the [BLA](#) in practice, we will construct a periodic signal with the same [PSD](#) and [PDF](#) as the stochastic noise signal and use this signal to determine the [BLA](#). A good candidate signal to implement this idea is the so-called random-phase multisine signal. A random-phase multisine is defined in the time domain as the sum of F commensurate tones

$$u(t) = \sum_{k=1}^F A_k \sin(2\pi k f_0 t + \varphi_k) \quad (2.12)$$

where f_0 is the base tone of the multisine. A_k and φ_k are, respectively, the amplitude and phase of the k^{th} tone. Choosing the amplitudes A_k deterministically sets the [PSD](#) of the signal. The phases φ_k set the [PDF](#). They are the realisation of a random process that sets the phases of one realisation of the multisine. For one such a phase realisation, the multisine is a deterministic, periodic signal. It has fixed values for the amplitudes A_k and phases φ_k . When all realisations are taken together, a random signal is created that belongs to the class of the original random excitation (Figure [2.3](#)). For a random-phase

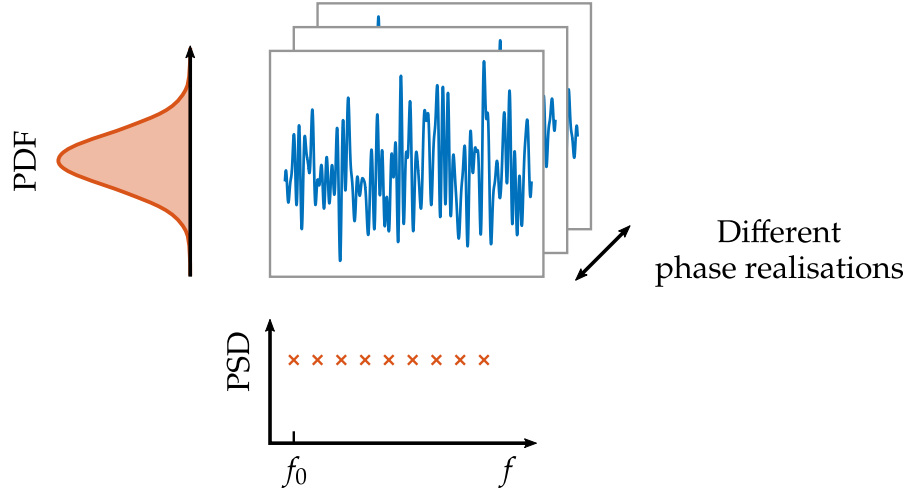


Figure 2.3: Random-phase multisines have a Gaussian PDF with a discrete PSD. Different phase realisations are used to emulate a random signal.

multisine, the phases φ_k are selected independently from a uniform distribution $[0, 2\pi)$ such that the multisine realisations resemble filtered white Gaussian noise. Independent controllability of both the amplitudes and the phases is important to efficiently mimic the PSD and the PDF of the original stochastic signals. After all, the same BLA is obtained for different signals that share a common PSD and PDF. Ideally, an infinite number of tones should be present in the multisine to emulate a continuous PSD. Based on our practical experience, a good rule of thumb is to have at least 30 tones to obtain satisfactory results. Another attractive property of the multisine signal is that it is a periodic signal. The periodicity implies that a proper experimental setup ensures that spectral leakage can be avoided completely in the steady-state input and output spectra. This simplifies the BLA estimation techniques. Furthermore, the distortion term also becomes periodic with the same period (PISPO) which additionally ensures that it can be separated from the stochastic noise term.

2.3.1 The robust method

Over the years, several BLA estimation techniques for systems excited with these multisines have been developed. The most straightforward one is the so-called robust method. This method computes (2.7) by considering M independent phase realisations of the

multisine signal. The changing phases result in a different time-domain signal and, consequently in a different nonlinear behaviour. Great care needs to be taken to ensure that the changing phases do not alter the PDF of the multisine excitations. Otherwise, different BLAs are obtained. The method requires that M different steady-state inputs $u(t)$ and outputs $y(t)$ are obtained for these M independent phase realisations. Computing the average and the variance over these realisations is then sufficient to estimate the BLA and the combined variance of the noise and the nonlinear distortions. To compute the noise variance separately, one needs to perform a statistical noise analysis for every phase realisation. A natural way to do this is to retrieve P repetitions of the steady-state input-output spectra for a certain phase realisation. After all, re-application of a multisine with the same phase realisation does not change the nonlinear behaviour if the system is PISPO. Differences over these P spectra are exclusively caused by the noise source $n(t)$. It is then sufficient to compute the variance over these P spectra to compute the noise variance separately. The number P is also called the number of periods of the multisine.

Let us now put some mathematics behind the robust method to more explicitly understand how the methods works. Consider that the input is noise-free and that we have gathered input-output data for several multisine realisations M that contain multiple periods P (Figure). The output signal of the system, measured in steady-state when the system is excited by the m^{th} phase realisation and the p^{th} period of the multisine is indicated with $y^{[m,p]}(t)$. As the input signal is not corrupted by noise, there is no difference from one period to the other. Therefore, we only consider a single noise-free input signal for each realisation that we denote with $u^{[m]}(t)$. The spectra of the different input-output segments are obtained by means of the DFT

$$U^{[m]}(k) = \text{DFT}\{u^{[m]}(t)\} \quad Y^{[m,p]}(k) = \text{DFT}\{y^{[m,p]}(t)\} \quad (2.13)$$

Before calculating the BLA, the influence of the noise needs to be determined first by averaging the output spectrum over the P periods. The averaging results in the following estimates of the output sample mean $\hat{Y}^{[m]}$ and sample noise variance $\hat{\sigma}_n^{2[m]}$ in the frequency domain

$$\hat{Y}^{[m]}(k) = \frac{1}{P} \sum_{p=1}^P Y^{[m,p]}(k) \quad (2.14)$$

$$\hat{\sigma}_{Y,n}^{2[m]}(k) = \frac{1}{P(P-1)} \sum_{p=1}^P |Y^{[m,p]}(k) - \hat{Y}^{[m]}(k)|^2 \quad (2.15)$$

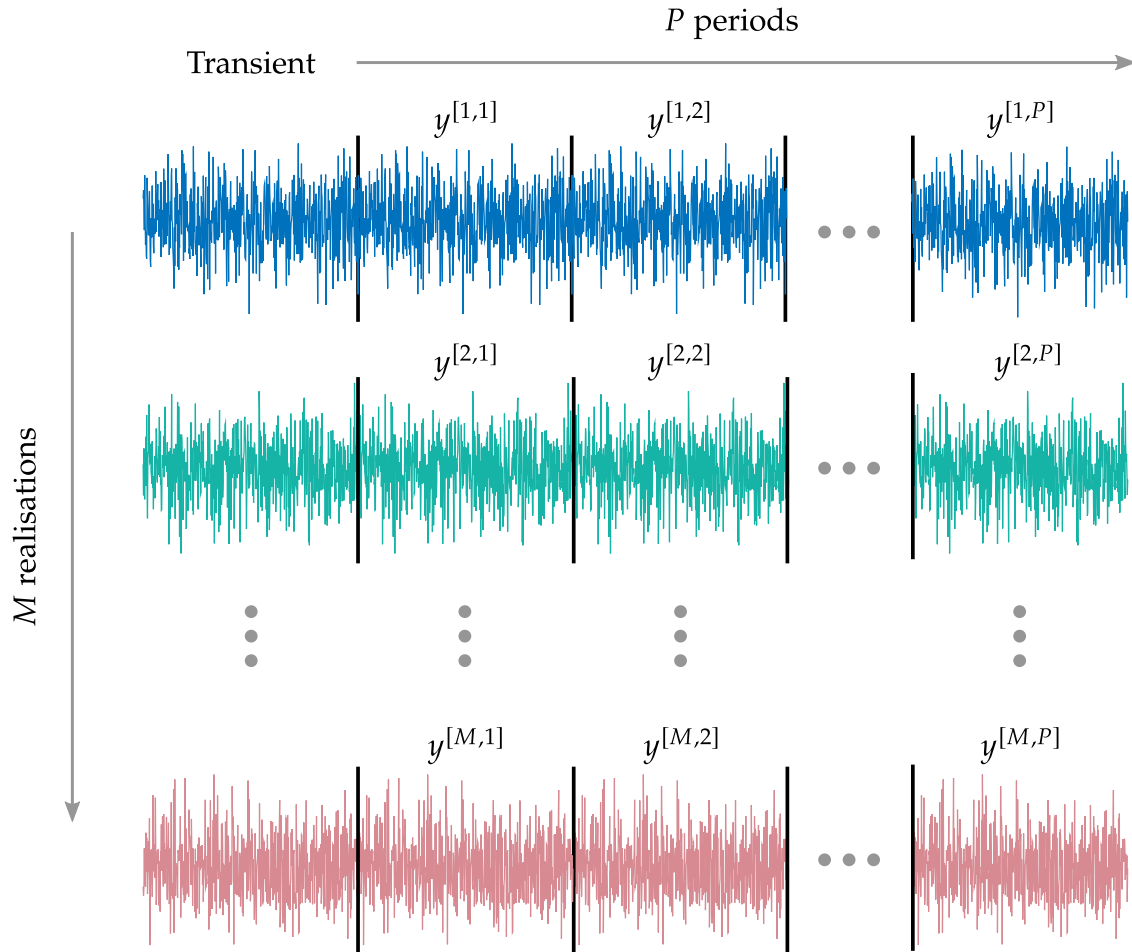


Figure 2.4: The robust method to calculate the *BLA* requires the acquisition of P consecutive periods of the steady-state response to M independent phase realisations of random-phase multisine excitations. During the different experiments only the phase realisation of the multisine excitation is varied. The amplitude profile and thus the *PSD* remains constant.

An additional factor $1/P$ is added in (2.15) to express that it is the sample noise variance of the *sample mean* and not of one single realisation.

Once the output sample mean is available, the **BLA** of the system is determined as the average frequency response of the system with respect to the phase realisations

$$G^{\text{BLA}}(j\omega_k) = \mathbb{E}_U \left\{ \frac{\hat{Y}^{[m]}(k)}{U^{[m]}(k)} \right\} \cong \frac{1}{M} \sum_{m=1}^M \frac{\hat{Y}^{[m]}(k)}{U^{[m]}(k)} \quad (2.16)$$

Note that the **BLA** can only be determined at the excited frequencies of the multisine, otherwise, a division by zero is performed. The approximation of the expected value will be accurate when the amount of phase realisations M is very large. In practice, we look to the total variance on the **BLA** estimate to determine whether more phase realisation are needed to obtain a more accurate estimate. The sample total variance $\hat{\sigma}_{G^{\text{BLA}}}^2$ and the sample noise variance $\hat{\sigma}_{G^{\text{BLA}},n}^2$ of the **BLA** estimate are calculated as

$$\begin{aligned} \hat{\sigma}_{G^{\text{BLA}}}^2(k) &= \frac{1}{M(M-1)} \sum_{m=1}^M \left| \frac{\hat{Y}^{[m]}(k)}{U^{[m]}(k)} - G^{\text{BLA}}(j\omega_k) \right|^2 \\ \hat{\sigma}_{G^{\text{BLA}},n}^2(k) &= \frac{1}{M^2} \sum_{m=1}^M \frac{\hat{\sigma}_{Y,n}^2(k)}{|U^{[m]}(k)|^2} \end{aligned} \quad (2.17)$$

Again, an additional factor M is included since we are dealing with the sample variance of the mean value.

In the case no output noise is present, the nonlinear distortion $D^{[m]}(k)$ introduced by the system as the response to a single phase realisation is defined as

$$D^{[m]}(k) = Y^{[m]}(k) - G^{\text{BLA}}(j\omega_k) U^{[m]}(k) \quad (2.18)$$

When the average behaviour of the distortion signal over many different phase realisations is considered, its noise-like properties appear:

- The average distortion at every frequency line is zero mean: $\mathbb{E}_U\{D(k)\} = 0$
- The distortion is uncorrelated to the input signal: $\mathbb{E}_U\{D(k) U^*(k)\} = 0$
- The average power of the distortion signal $\mathbb{E}_U\{D(k) D^*(k)\}$ is a continuous function of frequency

In the properties listed above, the expected value is always taken over the different phase realisations of the multisine.

If output noise is present, the power of the distortion can be determined if one considers the expected values of the sample total and noise variances in (2.17)

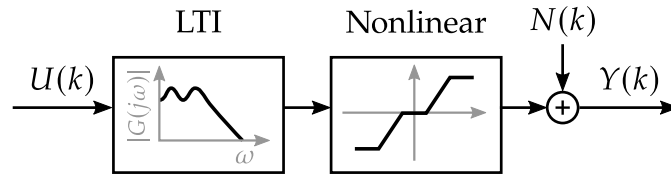
$$\begin{aligned}\mathbb{E}\{\hat{\sigma}_{G^{BLA}}^2\} &= \frac{\sigma_D^2(k)}{M|U_0(k)|^2} + \frac{\sigma_{Y,n}^2(k)}{MP|U_0(k)|^2} \\ \mathbb{E}\{\hat{\sigma}_{G^{BLA},n}^2(k)\} &= \frac{\sigma_{Y,n}^2(k)}{MP|U_0(k)|^2}\end{aligned}\quad (2.19)$$

with $\sigma_D^2(k) = \text{var}(D^{[m]}(k))$ is the variance of the nonlinear distortion of one experiment, $\sigma_{Y,n}^2(k) = \text{var}(N^{[m,p]}(k))$ is the output noise variance of one period, and $|U_0(k)|^2$ is the power of the input signal that is independent of the random phase realisation. From (2.19), it can be seen that the contribution of the output noise variance $\sigma_{Y,n}^2(k)$ is the same for $\mathbb{E}\{\hat{\sigma}_{G^{BLA}}^2\}$ and $\mathbb{E}\{\hat{\sigma}_{G^{BLA},n}^2(k)\}$. Therefore, an estimate of the sample variance of the nonlinear distortion can be obtained as

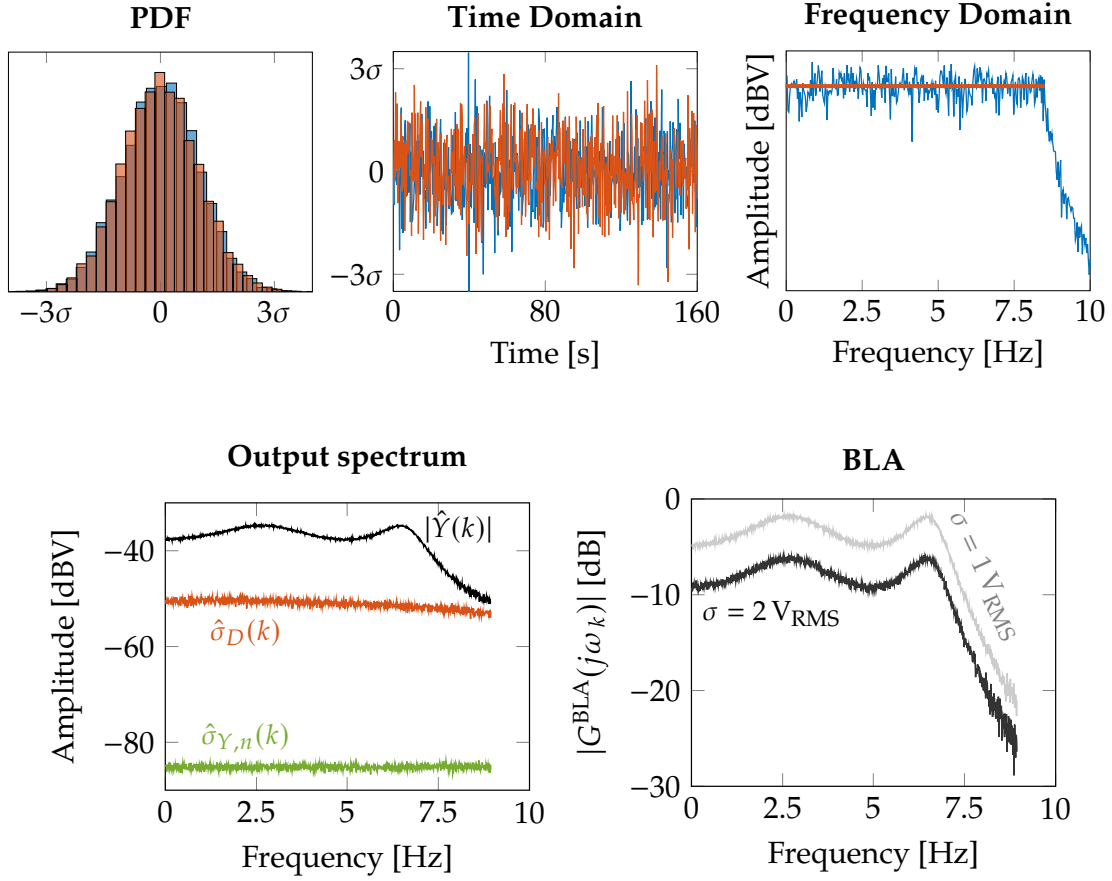
$$\hat{\sigma}_D^2(k) = \begin{cases} M\left(\hat{\sigma}_{G^{BLA}}^2 - \hat{\sigma}_{G^{BLA},n}^2\right)|U_0(k)|^2 & \hat{\sigma}_{G^{BLA}}^2 > \hat{\sigma}_{G^{BLA},n}^2 \\ 0 & \hat{\sigma}_{G^{BLA}}^2 \leq \hat{\sigma}_{G^{BLA},n}^2 \end{cases} \quad (2.20)$$

Example

The goal of this example is to showcase the capabilities of multisines for emulating filtered white Gaussian noise signals for estimating the BLA. Therefore, we consider the nonlinear system displayed below, which is a cascade of a linear dynamic Chebychev filter and a nonlinear static function that includes a deadzone and saturation. Furthermore, the output of the cascade is perturbed by white Gaussian noise with a standard deviation equal to 0.01 V. Suppose that we would want to derive the BLA for a filtered white Gaussian noise



excitation (—) with a standard deviation σ equal to 1 V (see figure below). Instead of using the original signal, we emulate the behaviour using a random-phase multisine (—) with $F = 1000$ tones. Comparing the PDF and the time-domain waveforms shows that the random-phase multisine indeed resembles the original Gaussian noise excitation. The



main difference becomes clear in the frequency domain where the multisine has complete control over the [PSD](#).

$M = 100$ phase realisations and $P = 2$ steady-state periods of the random-phase multisine are acquired using a time-domain simulator with a fixed time-step. One additional period is used during the simulation to ensure that the transient is completely damped out and steady-state is reached. The robust method is used to identify the [BLA](#) and estimates for the standard deviation of the noise $\hat{\sigma}_{Y,n}(k)$ and nonlinear distortion contributions $\hat{\sigma}_D(k)$ in the output spectrum: A significant portion of the output spectrum is clearly perturbed by the nonlinear distortion, which in turn results in a 'noisy' estimate of the mean output $\hat{Y}(k)$ and the [BLA](#). The noise on the other hand has, in this example, a marginal contribution in the output spectrum. Additionally, we show the influence of the input amplitude of the resulting [BLA](#) by doubling the input [RMS](#) value σ . The [BLA](#) clearly shows a significant reduction of the overall gain. This gain reduction is caused by the static nonlinear function that saturates the power-elevated signal relatively more compared to the original power

level. This example showed that the BLA is an effective tool to analyse the approximative linear behaviour of a dynamic nonlinear system, together with a characterisation of the noise and distortion levels.

2.3.2 Tailoring multisines for the detection of nonlinearities

The greatest benefit of working with periodic signals is that they can be modified easily to gain more insight into the nonlinear distortion generated by the nonlinear system. By carefully choosing the non-excited lines in the excitation signal, the distortion generated by a system can be determined just by inspection of the steady-state spectra without needing to calculate the BLA of the system.

An even nonlinearity in a system mixes the input signal an even amount of times, so an even amount of tones from the input signal are combined to obtain the even contributions. When the multisine excites only odd frequency bins, all even-order distortion only appears on even frequency bins, separating the even and odd nonlinear distortion contributions (Figure 2.5). This multisine, which excites only odd frequency bins ($A_k = 0$ for all even k) is called an *odd multisine*. Note that DC ($k = 0$) should not be excited since $k = 0$ is even.

The odd frequency bins in the output response contain the excited frequency lines and the odd-order nonlinear distortion. To determine the statistics of the odd-order distortion at an excited line, we can calculate the BLA using multiple phase realisations of the multisine and subtract the contribution due to the BLA as in (2.18). Alternatively, some of the odd frequencies can be left unexcited as well, to obtain detection lines in the multisine. These odd detection lines only contain contributions due to odd-order nonlinear distortion. To avoid disturbing the noise-like properties of the nonlinear distortion, the detection lines are chosen randomly out of groups of 4 excited lines in the odd multisine. The resulting odd multisine with randomly spaced detection lines is called a *random-odd multisine*.

Another advantage of using an odd multisine is that there is less uncertainty on the estimate of the BLA, since the even-order distortion falls on the non-excited even frequency lines. This can reduce the amount of phase realisations needed to obtain a low uncertainty. Note that these special multisines are only useful for base-band systems. In narrow-band bandpass multisines, the even and odd nonlinear contributions are automatically split.

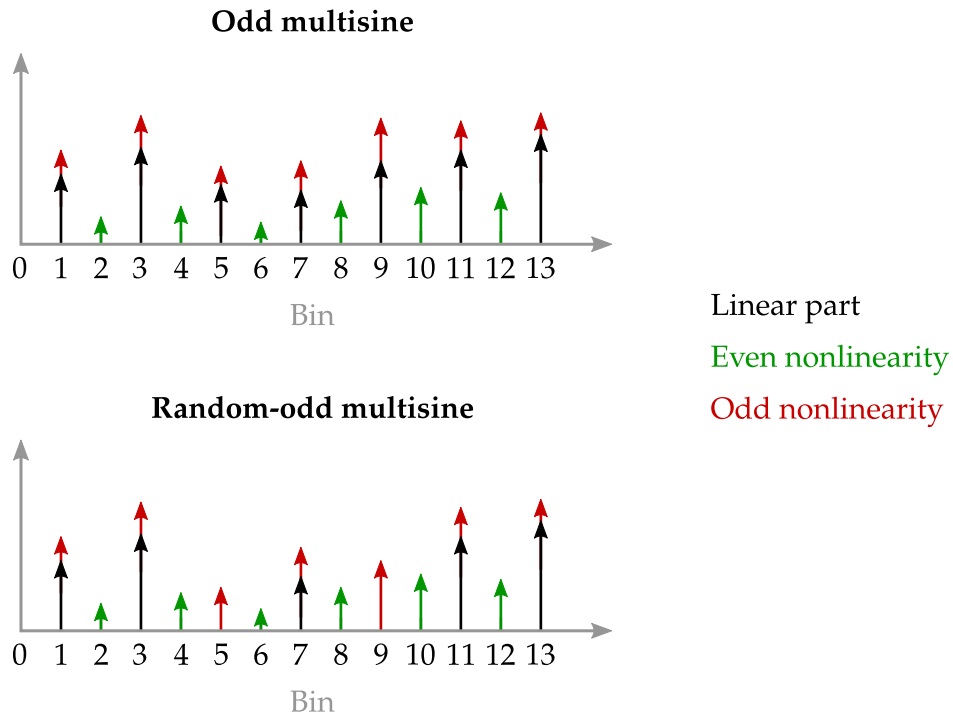


Figure 2.5: Cleverly choosing the excited frequencies in a multisine enables the detection and separation of the even- and odd-order distortion.

Detection lines can still be used there, but only for quick inspection of the in-band odd-order distortion.

2.4 Back to the case study

Now that we have established how the [BLA](#) theory works, we can use it to linearly approximate the behaviour of our nonlinear power amplifier. Section [1.4](#) describes the measurement campaign that applied a modulated signal to the amplifier at different power levels. This modulated signal is actually a random-odd multisine that excites 66 frequency tones within the 3 MHz bandwidth. Figure [1.8](#) clearly shows the presence of the detection lines within the excitation bandwidth.

The robust method is used to determine the [BLA](#) as a function of the applied input power. For each power level, $P = 10$ consecutive periods to $M = 16$ independent random-phase

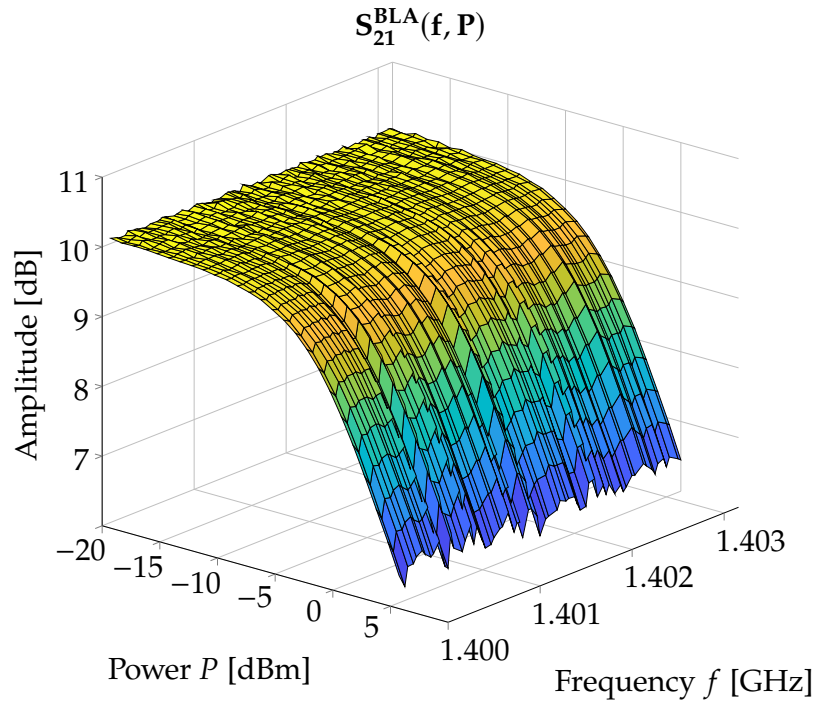


Figure 2.6: The forward gain S_{12}^{BLA} as a function of power clearly shows compression.

realisations is acquired with the measurement setup. The first period is not included in the calculation of the BLA as this period contains transient phenomena that violate the required steady-state assumption. The resulting forward gain S_{12}^{BLA} as a function of the modulation frequency f and the input power P is shown in Figure 2.6. As the modulation bandwidth is fairly narrow, the power amplifier behaves as a quasi-static device, which results in an almost constant BLA as a function of the modulation frequency. Starting from a power level of -5 dBm (-1 dB compression point), the power amplifier is pushed into its nonlinear regime and compression becomes clearly visible. The further the power amplifier is pushed into compression, the more dynamic variations appear in the BLA estimate. These variations are not caused by inherent behaviour of the amplifier, but are produced by the nonlinear distortion contributions to the BLA . These contributions add uncertainty to the BLA estimation that appears to be 'dynamic' variations around the main nonlinear trend. At the expense of an increased measurement time, they can be mitigated by performing more measurement with different phase realisations of the multisine excitation.

In addition to estimating the BLA , the robust method also quantifies the noise variance and the variance of the nonlinear distortions by averaging. Figure 2.7 displays how large

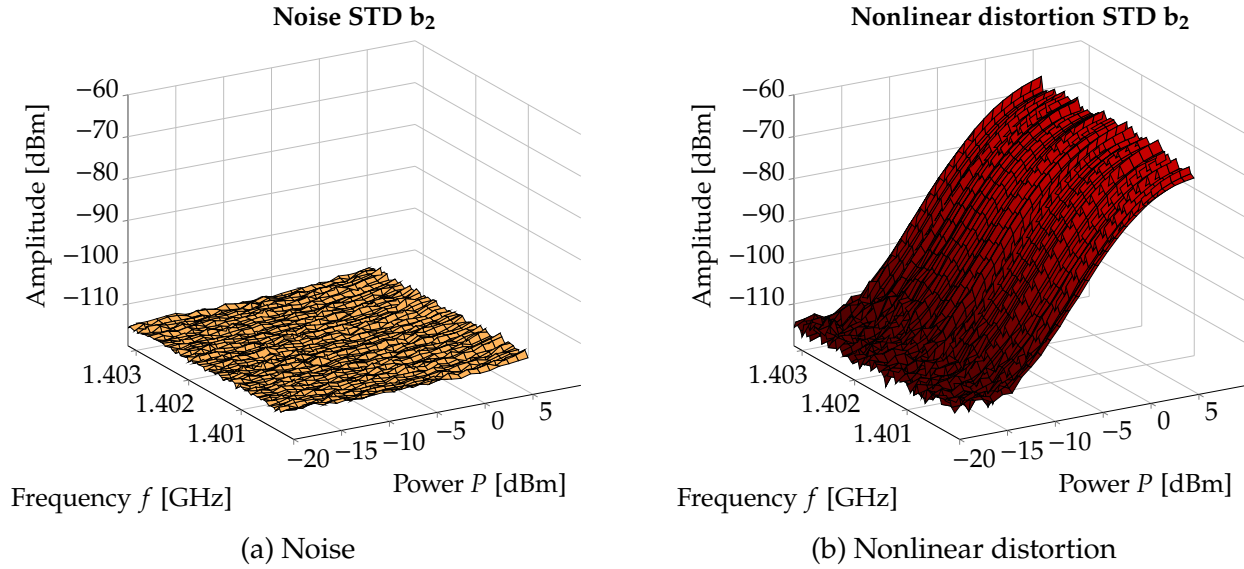


Figure 2.7: Estimated standard deviation (STD) of the noise and the nonlinear distortions of the b_2 spectrum.

the contribution of the noise and nonlinear distortions is in the b_2 spectrum. The noise contribution remains fairly constant as a function of the input amplitude at an average variance of -114 dBm. The contribution of the nonlinear distortions, on the other hand, increases significantly with increasing power levels. Similar to what was observed for the [BLA](#), the nonlinear distortion is also subject to compression at higher input power levels.

2.5 Summary

In this chapter, we have introduced the [Best Linear Approximation \(BLA\)](#), a powerful theory that enables you to describe the behaviour of both weakly and strongly nonlinear systems. The [BLA](#) is a black box model of the system, so no information about the internals of the system are required. Only knowledge about the input and output signals is required for the estimation of the [BLA](#).

The [BLA](#) is the linear model that describes the input-output behaviour of the system best in least-squares sense. It depends on both the system and the input signal properties ([PSD](#) and [PDF](#)). The residuals of the [BLA](#) approximation consist of measurement noise and

the distortion introduced by the system. This distortion has noise-like properties when the system is excited by a modulated signal: it is zero mean, is uncorrelated to the input signal, and has a smooth power spectrum. By combining these distortion properties with multisine excitations, it is possible to separate the distortion from the measurement noise and estimate their respective variances. In spite of these advantages, the [BLA](#) remains a linear approximation theory and can only predict the linearized behaviour of the nonlinear system at the excited frequencies. To fully describe the nonlinear behaviour of a system, a nonlinear model is needed, which is the topic of the next chapter.

Chapter 3

The Volterra theory of weakly nonlinear systems

Deciding which nonlinear model to use to describe the behaviour of the system is not a trivial task. As not one all-encompassing nonlinear theory exists, some models are better suited than others to represent particular aspects of the nonlinearity. Our case study of the nonlinear power amplifier illustrates this clearly. When we only want to determine its amplification, a linearized model such as the [BLA](#) is sufficient. However, a nonlinear model is necessary if we want, for example, to describe the distortions generated by the amplifier. Opting for an elementary model will yield a rough description of the system, but requires only limited computational complexity, while choosing an all-encompassing model will give a very accurate representation but necessitates very complex mathematics.

One approach to treat nonlinear systems makes use of functional series. Various representations are based on this very principle and spring from the need to identify an unknown nonlinear system. The basic idea is to use an infinite series of analytic functions that serves as a general approximant of any nonlinear system. For static nonlinearities, popular examples are the Taylor series, the Laurent series, and the trigonometric series. The Volterra theory extends this notion and uses analytic functional series to approximate the behaviour of a dynamic nonlinear system. This chapter clarifies the Volterra theory in a conceptual and natural way without formulating a strict mathematical environment. The emphasis lies on developing general intuition about this theory and how it can be used to interpret and study nonlinear behaviour.

3.1 The Volterra series

Consider the system h that explicitly relates the output $y(t)$ with the input $u(t)$ as defined in Section 1.1 and is repeated here for convenience

$$y(t) = h[u(t)] \quad (3.1)$$

Remember that the system h is a causal time-invariant nonlinear dynamic system. As a result, the output at a certain time instant t does not exclusively depends on the input value at that very instant t , but on the input value at every past instant of time. So the entire history of the excitation contributes to the value of the output at time instant t . While in general $u(t)$ and $y(t)$ should denote vector functions, as h may be a [Multiple-Input Multiple-Output \(MIMO\)](#) system, we confine ourselves to [Single-Input Single-Output \(SISO\)](#) systems to ease the notational complexity. Furthermore, remark that the explicit relationship between the input and output is an assumption we make about the behaviour of the system. This is not generally true for any nonlinear system as this, for example, does not include feedback mechanisms.

To intuitively derive the Volterra series, we will construct the output $y(t)$ for specific inputs $u(t)$ that are applied to the system h . Let us first start with determining the output when no excitation is applied, thus $u(t) = 0$. In general the output $y(t)$ may then be some constant denoted by g_0

$$y(t) = h[0] = g_0 \quad (3.2)$$

Pretend now that it would be possible to excite the system at some time instant τ_1 with a value $u(\tau_1)$. If $\delta(t)$ represents the Dirac impulse, then the resulting input becomes

$$u(t) = u(\tau_1) \delta(t - \tau_1) \quad (3.3)$$

The impact of applying this input at a certain time t will depend on the magnitude of the pulse $u(\tau_1)$, the time of occurrence τ_1 and the time instant t we are interested in. As we assume that the nonlinear system is time invariant, only the time difference $t - \tau_1$ is of importance. Therefore, we may represent the system's output at time instant t as some function of both the pulse amplitude $u(\tau_1)$ and time difference $t - \tau_1$

$$y(t) = h[u(\tau_1) \delta(t - \tau_1)] = L(u(\tau_1), t - \tau_1) \quad (3.4)$$

If we assume that the multivariate function L is continuously differentiable, then a Taylor series expansion of the input amplitude $u(\tau_1)$ around the origin results in

$$y(t) = g_0 + g_1(t - \tau_1) u(\tau_1) + g_2(t - \tau_1) u(\tau_1)^2 + g_3(t - \tau_1) u(\tau_1)^3 + \dots \quad (3.5)$$

Remark that the zeroth-order term in this expansion is identical to the response of the system when no input ($u(\tau_1) = 0$) is applied in (3.2).

Let us now extend the previous excitation by adding a second Dirac impulse at time instant τ_2 with a value $u(\tau_2)$. The input then consists out of two distinct pulses and may be formulated mathematically as

$$u(t) = u(\tau_1) \delta(t - \tau_1) + u(\tau_2) \delta(t - \tau_2) = \sum_{i=1}^2 u(\tau_i) \delta(t - \tau_i) \quad (3.6)$$

Just as before, the system's output at time instant t is now function of both amplitudes $u(\tau_1)$ and $u(\tau_2)$, as well as the two time differences $t - \tau_1$ and $t - \tau_2$. Therefore, the Taylor series expansion of the system as a function of $u(\tau_1)$ and $u(\tau_2)$ involves functions depending on the time differences $t - \tau_1$ and $t - \tau_2$

$$\begin{aligned} y(t) &= h \left[\sum_{i=1}^2 u(\tau_i) \delta(t - \tau_i) \right] \\ &\approx g_0 + g_1^1(t - \tau_1, t - \tau_2) u(\tau_1) + g_1^2(t - \tau_1, t - \tau_2) u(\tau_2) \\ &\quad + g_2^{11}(t - \tau_1, t - \tau_2) u(\tau_1)^2 + g_2^{12}(t - \tau_1, t - \tau_2) u(\tau_1) u(\tau_2) \\ &\quad + g_2^{21}(t - \tau_1, t - \tau_2) u(\tau_2) u(\tau_1) + g_2^{22}(t - \tau_1, t - \tau_2) u(\tau_2)^2 \\ &\quad + g_3^{111}(t - \tau_1, t - \tau_2) u(\tau_1)^3 + \dots + g_3^{222}(t - \tau_1, t - \tau_2) u(\tau_2)^3 + \dots \end{aligned} \quad (3.7)$$

This general expression of the Taylor series is quite involved and requires further adaptation to be compatible with certain properties of the applied excitation and system. First of all, we have to retrieve a relation identical to (3.5) when the amplitude of either one of the two pulses is made zero. To accommodate this, all functions in (3.7) that have identical top indices are only a function of the time difference corresponding with that particular

index

$$\begin{aligned}
y(t) \approx & g_0 + g_1^1(t - \tau_1)u(\tau_1) + g_1^2(t - \tau_2)u(\tau_2) \\
& + g_2^{11}(t - \tau_1)u(\tau_1)^2 + g_2^{12}(t - \tau_1, t - \tau_2)u(\tau_1)u(\tau_2) \\
& + g_2^{21}(t - \tau_1, t - \tau_2)u(\tau_2)u(\tau_1) + g_2^{22}(t - \tau_2)u(\tau_2)^2 \\
& + g_3^{111}(t - \tau_1)u(\tau_1)^3 + g_3^{112}(t - \tau_1, t - \tau_2)u(\tau_1)^2u(\tau_2) + \dots
\end{aligned} \tag{3.8}$$

We can furthermore apply a notational adaptation by interpreting all expressions h of some order n , marked by the lower index, to have n arguments of which some may be identical. This means we impose the following changes

$$\begin{aligned}
g_2^{11}(t - \tau_1) &= g_2^{11}(t - \tau_1, t - \tau_1) \\
g_2^{22}(t - \tau_2) &= g_2^{11}(t - \tau_2, t - \tau_2) \\
g_3^{111}(t - \tau_1) &= g_3^{111}(t - \tau_1, t - \tau_1, t - \tau_1) \\
g_2^{112}(t - \tau_1, t - \tau_2) &= g_2^{11}(t - \tau_1, t - \tau_1, t - \tau_2) \\
&\vdots
\end{aligned} \tag{3.9}$$

Since the system is assumed time invariant, the impact of one impulse shifted in time results in an identical output except for the time shift. Likewise, the response to a combination of pulses only depends on the relative timing of the pulses. In the special case these impulses coincide, we find that all the expressions h of some order n should stem from one and the same function.

$$\begin{aligned}
g_0 &\equiv h_0 \\
g_1^1(t_1) &= g_1^2(t_1) \equiv h_1(t_1) \\
g_2^{11}(t_1, t_2) &= g_2^{12}(t_1, t_2) = g_2^{21}(t_1, t_2) = g_2^{22}(t_1, t_2) \equiv h_2(t_1, t_2) \\
g_3^{111}(t_1, t_2, t_3) &= g_2^{112}(t_1, t_2, t_3) = \dots g_2^{222}(t_1, t_2, t_3) \equiv h_3(t_1, t_2, t_3) \\
&\vdots
\end{aligned} \tag{3.10}$$

Remark that in general the functions $h_n(t_1, \dots, t_n)$ do not need to be symmetric (invariant under all permutations of their variables).

Now that we have established all these different adaptations, we can finally rewrite (3.7) as follows

$$\begin{aligned}
 y(t) = & h_0 + \sum_{i=1}^2 h_1(t - \tau_i) u(\tau_i) + \sum_{i=1}^2 \sum_{j=1}^2 h_2(t - \tau_i, t - \tau_j) u(\tau_i) u(\tau_j) + \\
 & \sum_{i=1}^2 \sum_{j=1}^2 \sum_{k=1}^2 h_3(t - \tau_i, t - \tau_j, t - \tau_k) u(\tau_i) u(\tau_j) u(\tau_k) + \dots
 \end{aligned} \tag{3.11}$$

If we add more impulses at certain time instants, it suffices to adapt the upper bound of summation in (3.11) from 2 to the total number of considered Dirac impulses.

The expression in (3.11) only holds for signals that consist of discrete pulses. If we want to find an equivalent relationship for a continuous signal, we need to approximate the signal with a staircase curve and let the step size of the stairs become infinitesimally small in the limit. As a result of this operation, we obtain for the signal

$$u(t) = \int_{-\infty}^{+\infty} u(\tau) \delta(t - \tau) d\tau \tag{3.12}$$

Consequently, it is just as if the summation sign in (3.6) has become an integral in the limit. Likewise, it can be shown that the summations in (3.11) become integrals. The continuous-time equivalent of (3.11) then turns into

$$\begin{aligned}
 y(t) = & h_0 + \int_{-\infty}^{+\infty} h_1(t - \tau_i) u(\tau_i) d\tau_i + \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} h_2(t - \tau_i, t - \tau_j) u(\tau_i) u(\tau_j) d\tau_i d\tau_j + \\
 & \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} h_3(t - \tau_i, t - \tau_j, t - \tau_k) u(\tau_i) u(\tau_j) u(\tau_k) d\tau_i d\tau_j d\tau_k + \dots
 \end{aligned} \tag{3.13}$$

By modifying the integration variables, we obtain the Volterra series representation of the nonlinear system h as it is usually formulated in literature

$$\begin{aligned}
 y(t) = & h_0 + \int_{-\infty}^{+\infty} h_1(\tau_1) u(t - \tau_1) d\tau_1 + \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} h_2(\tau_1, \tau_2) u(t - \tau_1) u(t - \tau_2) d\tau_1 d\tau_2 + \dots + \\
 & \int_{-\infty}^{+\infty} \dots \int_{-\infty}^{+\infty} h_n(\tau_1, \dots, \tau_n) u(t - \tau_1) \dots u(t - \tau_n) d\tau_1 \dots d\tau_n + \dots
 \end{aligned} \tag{3.14}$$

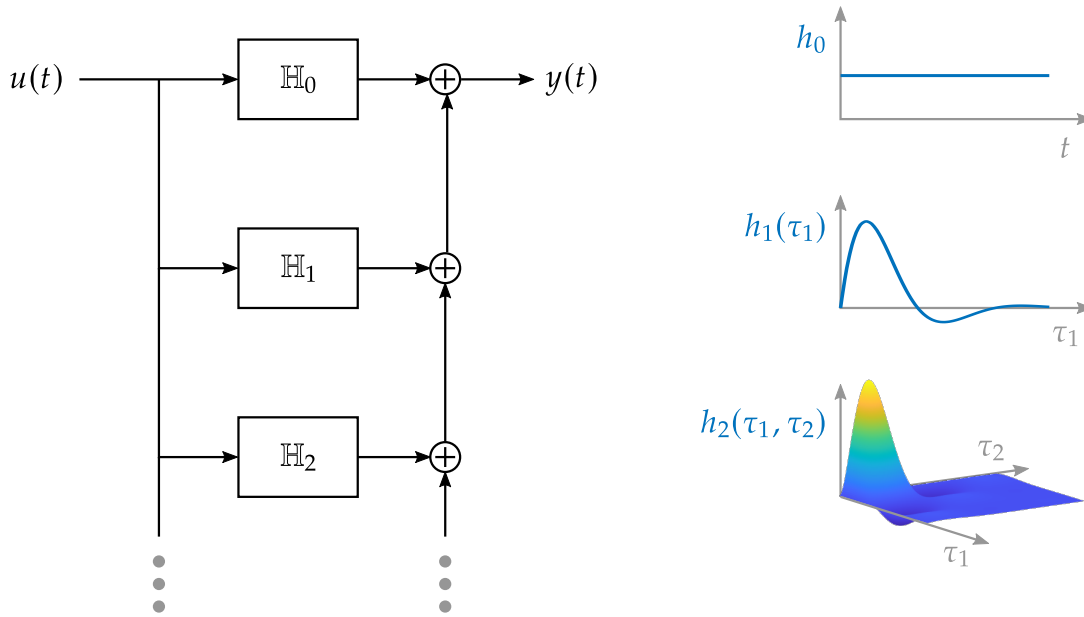


Figure 3.1: The Volterra series description describes the output of the system as a parallel cascade of Volterra operators of increasing order.

The series were named after the Italian mathematician V. Volterra, who studied it for the first time in the 1880s as a generalization of the Taylor series expansion of a nonlinear system.

In the Volterra series description, the output of such a weakly nonlinear system is described as a sum of terms (Figure 3.1). Each term can be seen as an operator acting on the input

$$y(t) = \sum_{n=0}^{\infty} \mathbb{H}_n[u(t)] \quad (3.15)$$

where $\mathbb{H}_1, \mathbb{H}_2, \dots, \mathbb{H}_n$ represent the Volterra operator of order 1, 2, $\dots$, n respectively. Together these Volterra kernels fully describe the nonlinear behaviour of the system. Each Volterra operator $\mathbb{H}_n$ consists of a n time-shifted multiplications of the input $u(t)$ and the function $h_n(\tau_1, \tau_2, \dots, \tau_n)$. The function is called the n^{th} -order Volterra kernel and can be considered as an n^{th} -order impulse response. Furthermore, (3.14) indicates that the output signal of an n^{th} -order Volterra operator $\mathbb{H}_n$ is computed by taking the n -dimensional convolution of the Volterra kernel h_n with the input signal.

The Volterra series can be viewed as a direct extension of linear systems theory to weakly nonlinear systems. When a signal is excited by a signal with very low input amplitude,

then the output can be described accurately by only taking into account the first-order behaviour of the Volterra series. Notice that the first-order Volterra operator corresponds to the well-known convolution formula for linear time-invariant systems. Therefore, Volterra series are a straightforward generalization of the linear case. We emphasize this conclusion as it implies that the concepts mastered for linear systems can be adapted for nonlinear systems. When the input amplitude increases, then a substantial part of the output signal is caused by nonlinear effects. The higher the input amplitude, the more higher-order effects occur, which in turn requires more higher-order Volterra operators to adequately model the system.

3.2 Properties of the Volterra series

Now that we have formally introduced the Volterra series description of a nonlinear system, we can take a look into some important properties of this description and how to enforce some of these properties.

3.2.1 Convergence

As the Volterra operators $\mathbb{H}_n$ are n -dimensional convolutions, they can describe nonlinear systems that exhibit memory effects. A Volterra series is thus nothing else but a power series with memory. This can be illustrated by observing the power of the gain factor A when exciting the system with an input equal to $A u(t)$. From (3.15), we immediately obtain

$$y(t) = \sum_{n=0}^{\infty} \mathbb{H}_n[A u(t)] = \sum_{n=0}^{\infty} A^n \mathbb{H}_n[u(t)] \quad (3.16)$$

Furthermore, if the system described by the Volterra series is static then

$$h_n(\tau_1, \tau_2, \dots, \tau_n) = 0 \quad \text{for any } \tau_j > 0 \text{ with } j = 1, 2, \dots, n \quad (3.17)$$

and the Volterra series reduces to an ordinary Taylor series

$$y(t) = \sum_{n=0}^{\infty} h_n(0, 0, \dots, 0) u(t)^n \quad (3.18)$$

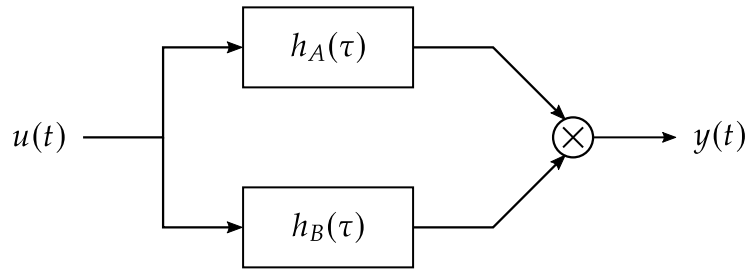


Figure 3.2: Example of a system with a non-symmetric second-order kernel.

The similarity between the Volterra and power series also implies they share the same kind of limitations associated with the applicability of Volterra series to nonlinear problems. Just as an ordinary power series, convergence problems may occur. When the input amplitude increases, the higher-order terms in the Volterra series increase relatively more than the lower-order ones. As a result, the Volterra series can start to diverge at a certain input amplitude. We assume in the remainder of this text that the signals that are applied to the systems under consideration are small enough such that the Volterra series converges. Furthermore, the system under test should be a so-called weakly nonlinear system that has a convergence radius different from zero. There exist strongly nonlinear systems, such as an ideal clipper or ideal comparator, that cannot be described at all with a converging Volterra series. For such systems, approximating the nonlinear behaviour of the system with a Volterra series is simply not possible.

3.2.2 Symmetrization and uniqueness

So far, we treated the Volterra kernels as abstract functions that are generally applicable for the description of nonlinear systems. However, if we want to introduce additional properties, we will have to impose structure on the general Volterra kernels. One of these properties is the uniqueness of the Volterra description. We will show in this section that symmetry of the Volterra kernels is essential to obtain this uniqueness.

In general, Volterra kernels are not necessarily symmetric. Let us consider the system shown in Figure 3.2 to illustrate this point. The system consists of two linear subsystems in parallel that act on the same input $u(t)$. Each linear subsystem is characterized by its respective one-dimensional impulse response $h_A(t)$ and $h_B(t)$. The output of the system is obtained by multiplying the signals coming out of the linear subsystems. Mathematically,

the output $y(t)$ can be calculated as follows

$$\begin{aligned}
 y(t) &= \int_{-\infty}^{+\infty} h_A(\tau_1) u(t - \tau_1) d\tau_1 \int_{-\infty}^{+\infty} h_B(\tau_2) u(t - \tau_2) d\tau_2 \\
 &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} h_A(\tau_1) h_B(\tau_2) u(t - \tau_1) u(t - \tau_2) d\tau_1 d\tau_2
 \end{aligned} \tag{3.19}$$

This expression shows that the Volterra series of the system consists only of a second-order Volterra operator with the following kernel

$$h_2(\tau_1, \tau_2) = h_A(\tau_1) h_B(\tau_2) \tag{3.20}$$

A kernel is called symmetric if it is invariant to any permutation of its arguments. For a second-order kernel this boils down to

$$h_2(\tau_1, \tau_2) = h_2(\tau_2, \tau_1) \tag{3.21}$$

It is clear that the kernel of the example system in (3.20) can only be symmetric if the impulse responses $h_A(t)$ and $h_B(t)$ are identical, which in general is not the case.

Remark, however, that the computation of $y(t)$ in (3.19) requires the integration over the entire argument range of the kernel. As a consequence, interchanging the integration variables τ_1 and τ_2 will not alter the result. Because of this observation, we may replace the second-order kernel by a symmetric one (h_2^{sym}) without making an error in the output calculation

$$\begin{aligned}
 h_2^{\text{sym}}(\tau_1, \tau_2) &= \frac{1}{2} (h_2(\tau_1, \tau_2) + h_2(\tau_2, \tau_1)) \\
 &= \frac{1}{2} (h_A(\tau_1) h_B(\tau_2) + h_A(\tau_2) h_B(\tau_1))
 \end{aligned} \tag{3.22}$$

Clearly, the observation made above holds equally well for any operator in the Volterra series. The symmetrization procedure can therefore be generalized to the n^{th} -order kernel

$$h_n^{\text{sym}}(\tau_1, \tau_2, \dots, \tau_n) = \frac{1}{n!} \sum_{\sigma \in S_n} h_n(\tau_{\sigma(1)}, \tau_{\sigma(2)}, \dots, \tau_{\sigma(n)}) \tag{3.23}$$

where S_n denotes all possible permutations of n arguments.

The purpose of making the kernel symmetric is two-fold. First, there is the convenience that you can disregard the order of the arguments in the kernel. This leads to expressions that are easier to handle, and a reduction in the number of calculations that need to be performed, which in turn results in a lower computation time. The second reason is even more important. It can be proven that the Volterra series with symmetric kernels is unique for every nonlinear system. From our earlier discussed example in Figure 3.2, it became clear that this is not the case when we consider nonsymmetric kernels. This uniqueness property is essential for the unambiguous identification of a nonlinear system by means of a Volterra series. As we are free to opt for symmetric kernels, it is implicitly assumed in what follows that all kernels are symmetric.

3.2.3 Causality and stability

Imposing causality and stability of a system in a Volterra series context can be done by applying additional conditions to the Volterra kernels.

Although non-causal systems can be meaningful in some applications, no physical system exists for which the output depends on future inputs. This very causality principle therefore needs to be reflected in the Volterra series representation of such a nonlinear system. For this purpose, consider the n^{th} -order Volterra kernel

$$\mathbb{H}_n[u(t)] = \int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} h_n(\tau_1, \dots, \tau_n) u(t - \tau_1) \dots u(t - \tau_n) d\tau_1 \dots d\tau_n \quad (3.24)$$

Here, t represents the present time. As is clear from the expression above, whenever one of the integration variables τ_i is smaller than zero, the future of the input is being addressed. Since the input to the system $u(t)$ may take any value as time evolves, the only way the output at time t can remain independent of the input beyond t , is by forcing the kernel to be zero whenever τ_i is smaller than zero

$$h_n(\tau_1, \tau_2, \dots, \tau_n) = 0 \quad \text{if any } \tau_i < 0 \text{ with } i = 1, 2, \dots, n \quad (3.25)$$

Clearly, the same reasoning is valid for any Volterra operator, thus the same conclusion holds for every Volterra kernel h_n . Remark that this property is automatically fulfilled for systems without memory.

Establishing if a nonlinear system exhibits (un)stable behaviour is not an easy task as there exist many definitions of stability. We select the "bounded-input bounded-output" criterion to determine stability. As the name suggests, this criterion states that a system is stable when every bounded input $u(t)$ results in a bounded output $y(t)$. By bounded input and output, we mean that there exist some finite numbers M_u and M_y such that $|u(t)| < M_u$ and $|y(t)| < M_y$ for all t . We will now show that a system is stable if each Volterra kernel respects the following constraint

$$\int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} |h_n(\tau_1, \tau_2, \dots, \tau_n)| d\tau_1 d\tau_2 \dots d\tau_n < \infty \quad \text{with } n = 0, 1, \dots \quad (3.26)$$

Remark that this condition implies that the distant past and future of the system, if we ignore causality, need to be irrelevant. To put it metaphorically, the Volterra kernels should thus show signs of dementia and exhibit a fading memory to be considered stable.

Let us now proof that when the input $u(t)$ is bounded, the constraint (3.26) is satisfied, and the Volterra series is convergent, that then the output of the system remains bounded

$$\begin{aligned} |y(t)| &= \left| \sum_{n=0}^{\infty} \int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} h_n(\tau_1, \dots, \tau_n) u(t - \tau_1) \dots u(t - \tau_n) d\tau_1 \dots d\tau_n \right| \\ &\leq \sum_{n=0}^{\infty} \int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} |h_n(\tau_1, \dots, \tau_n)| |u(t - \tau_1)| \dots |u(t - \tau_n)| d\tau_1 \dots d\tau_n \\ &< \sum_{n=0}^{\infty} M_u^n \int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} |h_n(\tau_1, \dots, \tau_n)| d\tau_1 \dots d\tau_n = M_y < \infty \end{aligned} \quad (3.27)$$

While this proves that the constraint (3.26) is sufficient to ensure stability, examples can be constructed demonstrating it is not necessary. Only in the linear time-invariant case, it can be shown that (3.26) is both a necessary and sufficient condition for bounded-input bounded-output stability.

3.3 Transitioning to the frequency domain

Up to now, we studied the Volterra series exclusively in the time domain. This section investigates how to represent the Volterra series in the frequency domain and what the potential benefits could be. Just as the frequency-domain transformation converts the troublesome convolution formula of a linear time-invariant system in an easy to use algebraic expression, we hope that the same conversion applies to Volterra series and that the multiple convolutions can be circumvented. To express the Volterra series approach in the frequency domain we need to implement three consecutive steps

1. Use the multidimensional Fourier transform to transform the Volterra kernel to the frequency domain.
2. Introduce the so-called 'associated functions' to deduce the input-output relationship in the multidimensional Fourier space.
3. Project the multidimensional output space onto one dimension to compute all the contributions in the one-dimensional output spectrum.

3.3.1 The multidimensional Fourier transform

As the n^{th} -order Volterra kernel $h_n(\tau_1, \dots, \tau_n)$ is a function of several variables τ_i , it is not possible to directly apply the well-known one-dimensional Fourier transform. To make the transformation to the frequency domain still possible, it is therefore required to use the n -dimensional Fourier transform introduced by Wiener. Let us first consider the two-dimensional case, before generalizing to the n -dimensional one.

The two-dimensional Fourier transform of the Volterra kernel $h_2(\tau_1, \tau_2)$ applies the one-dimensional Fourier transform to each time variable, one at a time, and is therefore a conceptually straightforward extension of the one-dimensional case. Suppose that τ_2 is held constant in $h_2(\tau_1, \tau_2)$, the one-dimensional Fourier transform $X(j\omega_1, \tau_2)$ with respect to τ_1 is given by

$$X(j\omega_1, \tau_2) = \int_{-\infty}^{+\infty} h_2(\tau_1, \tau_2) e^{-j\omega_1\tau_1} d\tau_1 \quad (3.28)$$

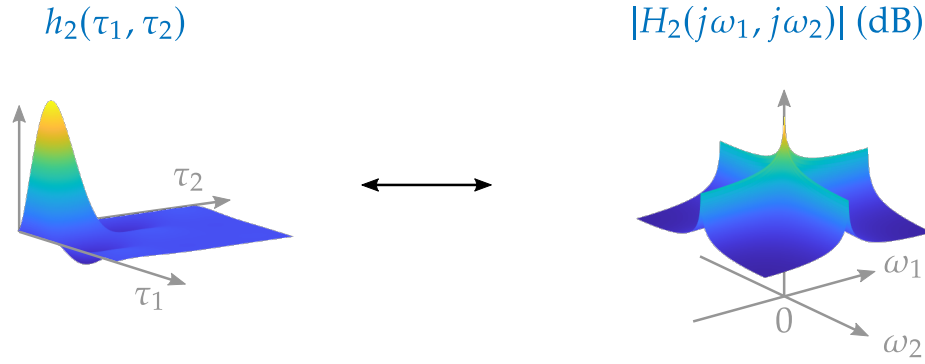


Figure 3.3: Time-domain kernel (left) and corresponding frequency-domain equivalent (right) of a stable second-order linear dynamic system cascaded by the quadratic term u^2 .

When the variable ω_1 is kept constant in this intermediate function, the one-dimensional Fourier transform of $X(j\omega_1, \tau_2)$ with respect to τ_2 is given by

$$H_2(j\omega_1, j\omega_2) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} h_2(\tau_1, \tau_2) e^{-j(\omega_1\tau_1 + \omega_2\tau_2)} d\tau_1 d\tau_2 \quad (3.29)$$

The function $H_2(j\omega_1, j\omega_2)$ is called the two-dimensional Fourier transform of the function $h_2(\tau_1, \tau_2)$ (Figure 3.3). The two-dimensional inverse transform of $H_2(j\omega_1, j\omega_2)$ is obtained in a similar manner

$$h_2(\tau_1, \tau_2) = \frac{1}{(2\pi)^2} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} H_2(j\omega_1, j\omega_2) e^{j(\omega_1\tau_1 + \omega_2\tau_2)} d\omega_1 d\omega_2 \quad (3.30)$$

The results obtained in the previous paragraph can be straightforwardly generalized to the n -dimensional case. The multidimensional Fourier transform $H_n(j\omega_1, \dots, j\omega_n)$ of the kernel $h_n(\tau_1, \dots, \tau_n)$ is given by

$$H_n(j\omega_1, \dots, j\omega_n) = \int_{-\infty}^{+\infty} \dots \int_{-\infty}^{+\infty} h_n(\tau_1, \dots, \tau_n) e^{-j(\omega_1\tau_1 + \dots + \omega_n\tau_n)} d\tau_1 \dots d\tau_n \quad (3.31)$$

and the inverse transformation is defined by

$$h_n(\tau_1, \dots, \tau_n) = \frac{1}{(2\pi)^n} \int_{-\infty}^{+\infty} \dots \int_{-\infty}^{+\infty} H_n(j\omega_1, \dots, j\omega_n) e^{j(\omega_1\tau_1 + \dots + \omega_n\tau_n)} d\omega_1 \dots d\omega_n \quad (3.32)$$

Equations (3.31) and (3.32) establish the multidimensional Fourier transform pair h_n and H_n . Remark that it is not automatically ensured that the multidimensional Fourier transform converges. To avoid convergence problems in the time-to-frequency transformation, it is therefore required that

$$\int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} |h_n(\tau_1, \dots, \tau_n)| d\tau_1 \dots d\tau_n < \infty \quad (3.33)$$

This constraint is automatically fulfilled if the system is stable, as it exactly corresponds to the sufficient condition in (3.26). It can furthermore be proven that the n^{th} -order Fourier transform is symmetric when its time-domain equivalent is symmetric. This is in fact a very natural result: a nonlinear operator does not discriminate between the different frequency arguments. An interesting property of the symmetric n^{th} -order Volterra kernel is that its complex conjugate can be obtained by changing the sign of the frequency arguments

$$H_n^*(j\omega_n, \dots, j\omega_n) = H_n(-j\omega_1, \dots, -j\omega_n) \quad (3.34)$$

Example

Similar to the linear case, the multidimensional Fourier transform $H_n(j\omega_1, \dots, j\omega_n)$ describes the magnitude and phase change a sinusoidal excitation undergoes when passing through the nonlinear system (see Section 1.1). While a linear system can only describe this change at the excitation frequency only, the Volterra series expands this notion and describes also the changes at multiples of the fundamental frequency (if only one tone is applied).

To back up this statement, let us consider a second- and third-order Volterra kernel and see what happens when we apply a one-tone steady-state excitation $u(t) = 2A \cos(\omega t) = Ae^{j\omega t} + Ae^{-j\omega t}$. For the second-order kernel h_2 , we obtain the following output to this excitation

$$\begin{aligned} y(t) &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} h_2(\tau_1, \tau_2) u(t - \tau_1) u(t - \tau_2) d\tau_1 d\tau_2 \\ &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} h_2(\tau_1, \tau_2) (Ae^{j\omega(t-\tau_1)} + Ae^{-j\omega(t-\tau_1)}) (Ae^{j\omega(t-\tau_2)} + Ae^{-j\omega(t-\tau_2)}) d\tau_1 d\tau_2 \end{aligned} \quad (3.35)$$

The expression above can be expressed as the summation of four separate integrals

$$\begin{aligned}
 y(t) = & A^2 e^{2j\omega t} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} h_2(\tau_1, \tau_2) e^{-j(\omega\tau_1 + \omega\tau_2)} d\tau_1 d\tau_2 \\
 & + A^2 e^{j0t} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} h_2(\tau_1, \tau_2) e^{-j(\omega\tau_1 - \omega\tau_2)} d\tau_1 d\tau_2 \\
 & + A^2 e^{j0t} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} h_2(\tau_1, \tau_2) e^{-j(-\omega\tau_1 + \omega\tau_2)} d\tau_1 d\tau_2 \\
 & + A^2 e^{-2j\omega t} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} h_2(\tau_1, \tau_2) e^{-j(-\omega\tau_1 - \omega\tau_2)} d\tau_1 d\tau_2
 \end{aligned} \tag{3.36}$$

Each one of these terms can be rewritten in terms of H_2 with differing frequency arguments

$$\begin{aligned}
 y(t) = & A^2 H_2(j\omega, j\omega) e^{2j\omega t} + A^2 H_2(-j\omega, -j\omega) e^{-2j\omega t} \\
 & + A^2 H_2(j\omega, -j\omega) + A^2 H_2(-j\omega, j\omega)
 \end{aligned} \tag{3.37}$$

The first two and the last two terms are complex conjugates of each other, which means the output reduces to the final result

$$y(t) = 2A^2 \operatorname{Re}\{H_2(j\omega, j\omega) e^{2j\omega t}\} + 2A^2 \operatorname{Re}\{H_2(j\omega, -j\omega)\} \tag{3.38}$$

The result above shows that the response of a second-order Volterra system to a sinusoidal excitation is a DC term and a sinusoid at the double frequency that are shaped by H_2 . Notice the similarity with the results obtained for the static nonlinear function $y = u^2$ in section 1.2. The main difference is that the output coefficients of the static nonlinearity do not exhibit frequency-dependent behaviour, while in the Volterra case $H_2(j\omega_1, j\omega_2)$ includes dynamic effects.

The same line of thought can be applied to the output calculation of the third-order Volterra kernel. Instead of four terms in the second-order case, eight different terms in the integral need to be considered for the third-order case. The resulting output consists of the following contributions

$$\begin{aligned}
 y(t) = & A^3 H_3(j\omega, j\omega, j\omega) e^{j3\omega t} + A^3 H_3(-j\omega, -j\omega, -j\omega) e^{-j3\omega t} \\
 & + A^3 H_3(j\omega, j\omega, -j\omega) e^{j\omega t} + A^3 H_3(j\omega, -j\omega, j\omega) e^{j\omega t} \\
 & + A^3 H_3(-j\omega, j\omega, j\omega) e^{j\omega t} + A^3 H_3(j\omega, -j\omega, -j\omega) e^{-j\omega t} \\
 & + A^3 H_3(-j\omega, j\omega, -j\omega) e^{-j\omega t} + A^3 H_3(-j\omega, -j\omega, j\omega) e^{-j\omega t}
 \end{aligned} \tag{3.39}$$

Taking into account symmetry, it is possible to rewrite the expression above to

$$y(t) = 2A^3 \operatorname{Re}\{H_3(j\omega, j\omega, j\omega) e^{j3\omega t}\} + 6A^3 \operatorname{Re}\{H_3(j\omega, j\omega, -j\omega) e^{j\omega t}\} \quad (3.40)$$

We can draw the same conclusion as in the static nonlinear case that the response of a cubic nonlinearity to a one-tone excitation consists of contributions at the original excitation frequency and the third harmonic.

3.3.2 Multidimensional input-output relationship

To express the Volterra series in the frequency domain we need to modify its usual representation in (3.14). Instead of writing down the series directly, we now cast the different time-domain outputs of the Volterra operators in a multidimensional format. In this alternative representation, the Volterra series is written as

$$y(t) = y_0 + y_1(t) + y_2(t, t) + y_3(t, t, t) + \dots + y_n(t, \dots, t) + \dots \quad (3.41)$$

where $y_n(t, \dots, t)$ is called the associated function of order n . This associated function is defined as

$$y_n(t_1, t_2, \dots, t_n) = \int_{-\infty}^{+\infty} \dots \int_{-\infty}^{+\infty} h_n(\tau_1, \tau_2, \dots, \tau_n) u(t_1 - \tau_1) u(t_2 - \tau_2) \dots u(t_n - \tau_n) d\tau_1 d\tau_2 \dots d\tau_n \quad (3.42)$$

You can see that the representation with the associated functions is equivalent with the original Volterra series description in (3.14). The main advantage of these associated functions is that we can directly transform them to the multidimensional Fourier domain by means of the earlier introduced multidimensional Fourier transform. Let us showcase how this transformation functions by calculating the frequency-domain representation of the second-order associated function

$$\begin{aligned} Y_2(\omega_1, \omega_2) &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} y_2(t_1, t_2) e^{j\omega_1 t_1} e^{j\omega_2 t_2} dt_1 dt_2 \\ &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} h_2(\tau_1, \tau_2) u(t_1 - \tau_1) u(t_2 - \tau_2) e^{j\omega_1 t_1} e^{j\omega_2 t_2} d\tau_1 d\tau_2 dt_1 dt_2 \end{aligned} \quad (3.43)$$

The time-shifting property of the Fourier transform can be used to transform the part related to the input

$$\begin{aligned} \int_{-\infty}^{+\infty} u(t_1 - \tau_1) e^{j\omega_1 t_1} dt_1 &= U(\omega_1) e^{j\omega_1 \tau_1} \\ \int_{-\infty}^{+\infty} u(t_2 - \tau_2) e^{j\omega_2 t_2} dt_2 &= U(\omega_2) e^{j\omega_2 \tau_2} \end{aligned} \quad (3.44)$$

which results in

$$Y_2(\omega_1, \omega_2) = \left(\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} h_2(\tau_1, \tau_2) e^{j(\omega_1 \tau_1 + \omega_2 \tau_2)} d\tau_1 d\tau_2 \right) U(\omega_1) U(\omega_2) \quad (3.45)$$

Closer inspection of the equation above reveals that the remaining integrals correspond to the two-dimensional Fourier transform as defined in (3.29). Taking this observation into account results in a straightforward relationship between the input and output spectra of the system

$$Y_2(\omega_1, \omega_2) = H_2(j\omega_1, j\omega_2) U(\omega_1) U(\omega_2) \quad (3.46)$$

The outcome for the second-order associated function can be straightforwardly generalized to the n^{th} -order associated function. In general, we find for the multidimensional Fourier transformation

$$Y_n(\omega_1, \omega_2, \dots, \omega_n) = H_n(j\omega_1, j\omega_2, \dots, j\omega_n) U(\omega_1) U(\omega_2) \dots U(\omega_n) \quad (3.47)$$

Notice once again how this expression forms a straightforward generalization of the linear case. For the Volterra series, a multidimensional convolution in the time domain converts to elementary multiplications in the multidimensional frequency domain. In this context, it is useful to mention that $H_n(j\omega_1, \dots, j\omega_n)$ is also named the n^{th} -order transfer function, in analogy with the notion transfer function that is well-known in linear system theory.

3.3.3 The one-dimensional output spectrum

To obtain the output spectrum $Y(\omega)$, we still need to convert the multidimensional spectra $Y_n(\omega_1, \dots, \omega_n)$ as time and frequency are one-dimensional quantities in reality. To

do so, we can start from the inverse multidimensional transform of $Y_n(\omega_1, \dots, \omega_n)$ and equate all the arguments to t . By means of an example let us again consider the second-order Volterra kernel to exemplify this operation. Applying the inverse multidimensional Fourier transform to $Y_2(\omega_1, \omega_2)$ results in

$$y_2(t_1, t_2) = \frac{1}{(2\pi)^2} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} Y_2(\omega_1, \omega_2) e^{j\omega_1 t_1} e^{j\omega_2 t_2} d\omega_1 d\omega_2 \quad (3.48)$$

and equating the arguments $t = t_1 = t_2$ gives

$$y_2(t, t) = \frac{1}{(2\pi)^2} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} Y_2(\omega_1, \omega_2) e^{j(\omega_1 + \omega_2)t} d\omega_1 d\omega_2 \quad (3.49)$$

The expression above shows that the one-dimensional projection results in output contributions at the sum of two frequencies $\omega_1 + \omega_2$, as was established earlier in the examples. To express this frequency addition mechanism more explicitly we can perform a change of integration variables

$$\omega = \omega_1 + \omega_2 \quad \Rightarrow \quad \omega_2 = \omega - \omega_1 \quad \text{and} \quad d\omega_2 = d\omega \quad (3.50)$$

Applying this change results in

$$y_2(t, t) = \frac{1}{(2\pi)^2} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} Y_2(\omega_1, \omega - \omega_1) e^{j\omega t} d\omega_1 d\omega \quad (3.51)$$

Using this equation, we can now calculate the contribution of the second-order kernel to the output spectrum as

$$\begin{aligned} Y_2(\omega) &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} Y_2(\omega_1, \omega - \omega_1) d\omega_1 \\ y_2(t, t) &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} Y_2(\omega) e^{j\omega t} d\omega \end{aligned} \quad (3.52)$$

Again, these results can be generalized to the n^{th} -order Volterra kernel. To obtain the one-dimensional frequency-domain contributions to the output spectrum, all the possible

frequency combinations that add up to a specific frequency ω need to be summed together. Mathematically this boils down to calculating the following relationship

$$Y_n(\omega) = \frac{1}{(2\pi)^{n-1}} \int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} Y_n\left(\omega_1, \omega_2, \dots, \omega - \sum_{i=1}^{n-1} \omega_i\right) d\omega_1 \dots d\omega_{n-1} \quad (3.53)$$

with

$$\omega = \omega_1 + \omega_2 + \cdots + \omega_n \quad (3.54)$$

The actual output spectrum $Y(\omega)$ can then be found by summing all $Y_n(\omega)$ for $n = 0, 1, 2, \dots$. It is needless to say that the presence of the convolution integrals keeps the Volterra series in the frequency domain complicated. So the frequency-domain version of the Volterra series does not yield the same simplifications as in the linear case. The multiple convolutions present in the time-domain version are just converted to similar convolutions in the frequency domain.

3.3.4 The output spectrum for periodic inputs

The introduced frequency-domain Volterra series enables you to calculate the steady-state output spectrum of a nonlinear system in response to an arbitrary excitation $u(t)$. It was shown that several convolutions are required in both the time- and frequency-domain representation. This difficulty vanishes, however, when the output spectrum $Y(\omega)$ is determined for a discrete input spectrum $U(\omega)$. It is clear that for the nonlinear systems we consider, i.e. those who can be described by a converging Volterra series, the output spectrum is then discrete as well. As we are requiring that the input spectrum is discrete, we are restricting ourselves to periodic inputs.

A signal with a discrete spectrum is nothing else than a multisine excitation, which has been introduced earlier in Section 2.3. For convenience, the definition of a multisine excitation is repeated here in the complex exponential form, where each complex exponential has its own complex amplitude B_k , and frequency Ω_k

$$u(t) = \sum_{\substack{k=-N \\ k \neq 0}}^N B_k e^{j\Omega_k t} \quad (3.55)$$

To ensure that the multisine is a real signal we need to impose that $B_{-k} = B_k^*$ and $\Omega_{-k} = -\Omega_k$. Taking the Fourier transform of the multisine excitation results in a spectrum $U(\omega)$ that is discrete

$$U(\omega) = 2\pi \sum_{\substack{k=-N \\ k \neq 0}}^N B_k \delta(\omega - \Omega_k) \quad (3.56)$$

where δ represents the Dirac impulse.

Assume now that we have a nonlinear system described by a Volterra series, and that we excite this system with a signal that has a discrete spectrum. To find the output spectrum of the system, we first substitute (3.56) in (3.47) and obtain

$$\begin{aligned} Y_n(\omega_1, \dots, \omega_n) &= (2\pi)^n H_n(\omega_1, \dots, \omega_n) \left(\sum_{\substack{k_1=-N \\ k_1 \neq 0}}^N B_{k_1} \delta(\omega_1 - \Omega_{k_1}) \right) \dots \left(\sum_{\substack{k_n=-N \\ k_n \neq 0}}^N B_{k_n} \delta(\omega_n - \Omega_{k_n}) \right) \\ &= (2\pi)^n \sum_{k_1} \dots \sum_{k_n} H_n(\omega_1, \dots, \omega_n) B_{k_1} \dots B_{k_n} \delta(\omega_1 - \Omega_{k_1}) \dots \delta(\omega_n - \Omega_{k_n}) \end{aligned} \quad (3.57)$$

The output spectrum can then be obtained by summing over all the different Volterra kernels and apply for each kernel $n - 1$ convolutions to project the multidimensional Fourier space onto one dimension with $\omega = \omega_1 + \dots + \omega_n$

$$\begin{aligned} Y(\omega) &= 2\pi \sum_{n=0}^{\infty} \underbrace{\sum_{k_1} \dots \sum_{k_n}}_n B_{k_1} \dots B_{k_n} \underbrace{\int_{-\infty}^{+\infty} \dots \int_{-\infty}^{+\infty}}_{n-1} H_n \left(\omega_1, \dots, \omega - \sum_{i=1}^{n-1} \omega_i \right) \\ &\quad \times \delta(\omega_1 - \Omega_{k_1}) \dots \delta \left(\omega - \sum_{i=1}^{n-1} \omega_i - \Omega_{k_n} \right) d\omega_1 \dots d\omega_{n-1} \end{aligned} \quad (3.58)$$

We can apply the shifting property of the Dirac function, which is defined as

$$\int_{-\infty}^{+\infty} p(t) \delta(t - T) dt = p(T) \quad (3.59)$$

to simplify the expression of the output spectrum

$$Y(\omega) = 2\pi \sum_{n=0}^{\infty} \sum_{k_1} \dots \sum_{k_n} B_{k_1} \dots B_{k_n} H_n \left(\Omega_{k_1}, \Omega_{k_2}, \dots, \omega - \sum_{i=1}^{n-1} \Omega_{k_i} \right) \delta \left(\omega - \sum_{i=1}^n \Omega_{k_i} \right) \quad (3.60)$$

Example

Even though we are no longer troubled by integrals in (3.60), it is still far from trivial to determine the output spectrum. To have an idea about the complexity involved, let us compute the output contributions for a nonlinear system that is excited by a two-tone signal. The two-tone signal consists of four complex exponentials at frequencies Ω_1 , $-\Omega_1$, Ω_2 , and $-\Omega_2$. The Volterra series of the system is restricted to first- and second-order behaviour only. Working out a full example with one more order or tone would take more than one page to write down. In our example, we find for $Y(\omega)$

$$\begin{aligned}
Y(-2\Omega_2) &= H_2(-\Omega_2, -\Omega_2) B_2^* B_2^* \\
Y(-\Omega_2 - \Omega_1) &= H_2(-\Omega_2, -\Omega_1) B_2^* B_1^* + H_2(-\Omega_1, -\Omega_2) B_1^* B_2^* \\
Y(-2\Omega_1) &= H_2(-\Omega_1, -\Omega_1) B_1^* B_1^* \\
Y(-\Omega_2) &= H_1(-\Omega_2) B_2^* \\
Y(-\Omega_1) &= H_1(-\Omega_1) B_1^* \\
Y(-\Omega_2 + \Omega_1) &= H_2(-\Omega_2, \Omega_1) B_2^* B_1 + H_2(\Omega_1, -\Omega_2) B_1 B_2^* \\
Y(0) &= H_2(-\Omega_2, \Omega_2) B_2^* B_2 + H_2(-\Omega_1, \Omega_1) B_1^* B_1 + \\
&\quad H_2(\Omega_1, -\Omega_1) B_1 B_1^* + H_2(\Omega_2, -\Omega_2) B_2 B_2^* \\
Y(\Omega_2 - \Omega_1) &= H_2(-\Omega_1, \Omega_2) B_1^* B_2 + H_2(\Omega_2, -\Omega_1) B_2 B_1^* \\
Y(\Omega_1) &= H_1(\Omega_1) B_1 \\
Y(\Omega_2) &= H_1(\Omega_2) B_2 \\
Y(2\Omega_1) &= H_2(\Omega_1, \Omega_1) B_1 B_1 \\
Y(\Omega_1 + \Omega_2) &= H_2(\Omega_1, \Omega_2) B_1 B_2 + H_2(\Omega_2, \Omega_1) B_2 B_1 \\
Y(2\Omega_2) &= H_2(\Omega_2, \Omega_2) B_2 B_2
\end{aligned} \tag{3.61}$$

Notice again the frequency generation mechanism inherent to nonlinear systems. As seen earlier, these frequencies all occur at linear combinations of the four excitation frequencies.

It becomes clear from this basic example that the Volterra series is plagued by a combinatorial explosion of terms. If the system would have been linear, only the four complex terms corresponding with the first-order transfer function have to be considered. The presence of the second-order Volterra kernel requires 16 more complex terms to determine the output spectrum with only four excitation frequencies. In general, the number of terms that need to be considered for a n^{th} -order transfer function when the input is composed

of N real sine waves is given by

$$(2N)^n \quad (3.62)$$

This number can be nearly halved if we realize that the amplitude corresponding with a negative frequency is the complex conjugate of the amplitude associated with the same frequency taken positive. A more fundamental reduction is obtained if we assume that the transfer functions are symmetric. As a result of this symmetry, the arguments of a transfer function may be permuted without changing its value. This means we only have to consider those contributions for which these arguments form different combinations of the base frequencies.

3.4 Mitigating the combinatorial explosion with block-structured models

Despite all the nice features of the Volterra series description, obtaining a Volterra series for a practical system is far from trivial. The main cause for this inconvenience is that up to now we described the Volterra series in a non-parametric manner. For every combination of $(\tau_1, \dots, \tau_n)$ or $(\omega_1, \dots, \omega_n)$, we need to obtain a distinct term, which results in an exponential explosion of the Volterra terms with increasing order or number of time delays/tones. Luckily, this combinatorial explosion can be mitigated if we consider nonlinear block-structured models.

Block-structured nonlinear models are constructed starting from two basic building blocks: a [Linear Time-Invariant \(LTI\)](#) block and a [Static Nonlinear \(SNL\)](#) block. By connecting these blocks in series or parallel, a wide variety of block-oriented models can be constructed. Here, we restrict ourselves to the most popular block structures that are available in literature (Figure 3.4). By separating the nonlinear dynamic behaviour into [LTI](#) dynamics and [SNL](#) blocks, block-structured nonlinear models are quite simple to understand, easy to use, and result in a significant reduction in the number of Volterra terms to be estimated. Obviously, this reduction comes at a cost: it is accompanied by a lowered nonlinear modelling capability compared to the full Volterra series description.

In this section, we discuss how these block-structured models relate to the original Volterra series description. But before we dive into the comparison, we first need to adapt the

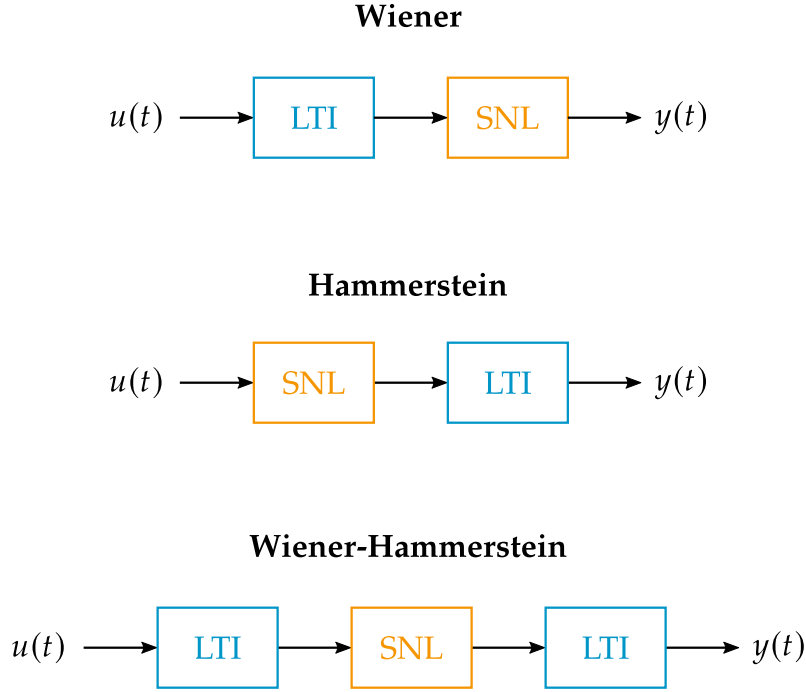


Figure 3.4: Single-branch structures that are obtained by using series connections of *LTI* and *SNL* blocks.

formulation of the general Volterra series description. Up to now, we considered that the Volterra series is continuous in time, has an infinite number of Volterra kernels, and considers an infinite number of time delays. To ensure that the Volterra series becomes feasible to work with in practice, we transition to the discrete-time formulation of the Volterra series

$$y(m) = \sum_{n=0}^{N_{\text{order}}} \left(\sum_{l_1=0}^{M_{\text{lag}}} \dots \sum_{l_n=0}^{M_{\text{lag}}} h_n(l_1, \dots, l_n) \prod_{p=1}^n u(m - l_p) \right) \quad (3.63)$$

where $u(m)$ and $y(m)$ are discrete-time signals that are obtained by equidistantly sampling the continuous-time signals $u(t)$ and $y(t)$, respectively. The maximal order of the Volterra series has been restricted to N_{order} , since for a weakly nonlinear system only a finite amount of Volterra kernels significantly contributes to the output. Furthermore, the Volterra series should have a fading memory to ensure stable behaviour (see Section 3.2.3). Due to this fading memory, we can also limit the number of time lags to M_{lag} , as we assume that time-delayed samples beyond $m - M_{\text{lag}}$ do not contribute anymore to the sample at time instance m .

3.4.1 Wiener

A first special case of the general Volterra series is the so-called Wiener model, which consists of a linear dynamic filter $h(m)$ followed by a static nonlinearity. If this nonlinearity can be written as a polynomial function

$$z = \sum_{n=0}^{N_{\text{order}}} a_n x^n \quad (3.64)$$

then the input-output relationship becomes

$$y_W(m) = \sum_{n=0}^{N_{\text{order}}} a_n \left(\sum_{l=0}^{M_{\text{lag}}} h(l) u(m-l) \right)^n \quad (3.65)$$

This relationship is obtained by first expressing the behaviour of the [LTI](#) block by means of a discrete-time convolution, followed by the application of the polynomial nonlinear function. The Wiener model is one of the most elementary ways to combine dynamic and nonlinear effects. Regarding model estimation it has unfortunately one undesirable attribute. The model output depends nonlinearly on the coefficients $h(l)$, which makes their estimation more difficult.

3.4.2 Hammerstein

Another elementary nonlinear model is the Hammerstein model, which is formed by a [SNL](#) block followed by an [LTI](#) filter $g(m)$. If we again consider the polynomial in (3.64) for the static nonlinear behaviour, then the model output of the Hammerstein model becomes

$$y_H(m) = \sum_{l=0}^{M_{\text{lag}}} g(l) \left(\sum_{n=0}^{N_{\text{order}}} a_n u^n(m-l) \right) \quad (3.66)$$

The Hammerstein model does have the desirable property of being linear in the parameters $g(l) a_n$, if they are considered as a two-dimensional array over l and n . By combining the different filter and power series coefficients into a two-dimensional array $\{a_{nl}\}$, we obtain the so-called *generalized* Hammerstein model

$$y_{GH}(m) = \sum_{n=0}^{N_{\text{order}}} \sum_{l=0}^{M_{\text{lag}}} a_{nl} u^n(m-l) \quad (3.67)$$

Notice that the Hammerstein model reduces the Volterra series model to its diagonal terms only, and completely neglects the cross-terms.

3.4.3 Wiener-Hammerstein

The Wiener and Hammerstein structures can be combined to form a cascade consisting of an [LTI](#) filter h , a [SNL](#) block, and another [LTI](#) filter g . The resulting Wiener-Hammerstein model structure results in the following model output

$$y_{\text{WH}}(m) = \sum_{l_2=0}^{M_{\text{lag}}} g(l_2) \sum_{n=0}^{N_{\text{order}}} a_n \times \left[\sum_{l_1=0}^{M_{\text{lag}}} h(l_1) u(m - l_1 - l_2) \right]^n \quad (3.68)$$

The Wiener-Hammerstein model structure is popular in literature and generally serves as a good approximant of the full-blown Volterra model. Remark that this model is still nonlinear in the parameters $h(l_1)$, which complicates the model estimation. Finding good initial values for the dynamic filter coefficients during the estimation is tricky, as it is not possible to fully decouple the influence of both [LTI](#) filters on the global behaviour.

3.4.4 Memory polynomials

The memory polynomial model is a very compact version of the Volterra series that has proven to be effective in linearizing power amplifiers. It is a specialized version of the generalized hammerstein model in (3.67) where the envelope of the input signal $|u|$ is introduced

$$y_{\text{MP}}(m) = \sum_{n=1}^{N_{\text{order}}} \sum_{l=0}^{M_{\text{lag}}} a_{nl} u(m - l) |u(m - l)|^{n-1} \quad (3.69)$$

This model structure can be easily converted into a parallel combination of M_{lag} polynomial functions, where each branch is applied to a delayed version of the input samples $u(m)$ (Figure 3.5). Due to the sectioning in several (delayed) nonlinear branches, the memory polynomials can be very efficiently implemented on an [Field-Programmable Gate Array \(FPGA\)](#), which makes this model structure very attractive for nonlinear modelling purposes in high-frequency applications. Furthermore, the model coefficients can be easily estimated with a linear least-squares solver as the model is linear in the parameters.

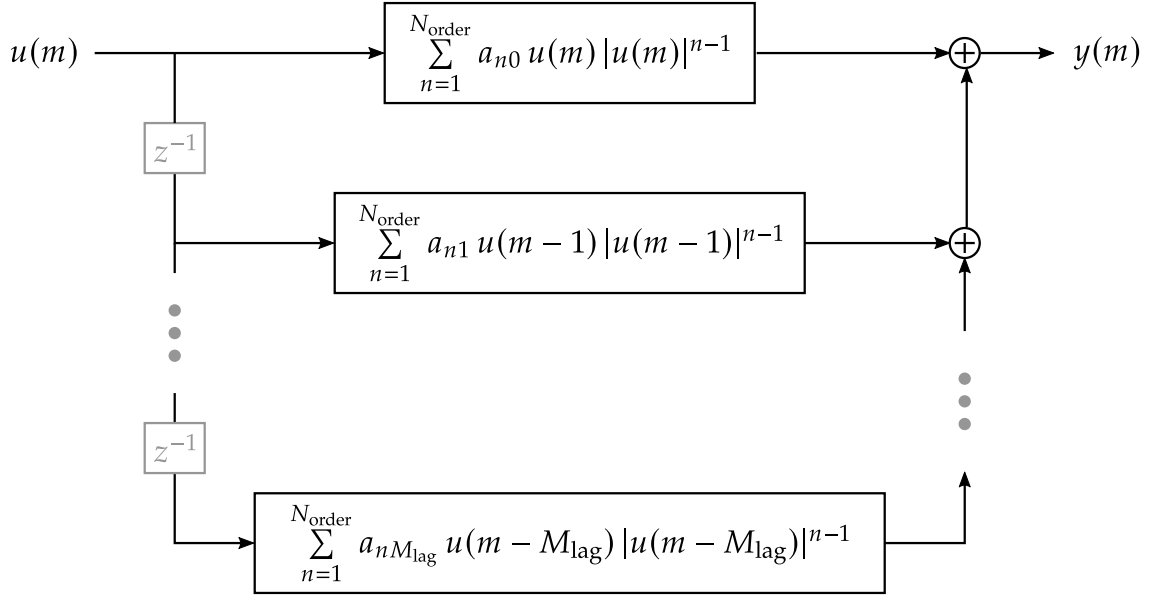


Figure 3.5: The memory polynomial structure sectioned into several nonlinear branches.

Given the input and output samples, it suffices to rewrite (3.69) in a matrix format

$$y_{\text{MP}}(m) = \mathbf{J}_{\text{MP}}(m) \cdot \mathbf{A} \quad (3.70)$$

where $\mathbf{J}_{\text{MP}}(m)$ and $\mathbf{A}$ are defined by

$$\mathbf{J}_{\text{MP}}(m) = \begin{bmatrix} u(m) \\ \vdots \\ u(m) |u(m)|^{N_{\text{order}}-1} \\ u(m-1) \\ \vdots \\ u(m-1) |u(m-1)|^{N_{\text{order}}-1} \\ \vdots \\ u(m-M_{\text{lag}}) |u(m-M_{\text{lag}})|^{N_{\text{order}}-1} \end{bmatrix}^T \quad (3.71)$$

$$\mathbf{A} = \begin{bmatrix} a_{10} & \dots & a_{N_{\text{order}}0} & a_{11} & \dots & a_{N_{\text{order}}1} & \dots & a_{N_{\text{order}}M_{\text{lag}}} \end{bmatrix}^T$$

Evaluating (3.70) for all values of m results in a set of linear equations that can be solved with a linear solver.

Over the years, several adaptations and improvements to the original memory polynomial model structure have been proposed to either increase the parameter parsimony or

improve the modelling capabilities. One approach to improve the performance is by augmenting the memory polynomial model with additional basis functions that introduce cross-terms with leading and lagging samples of the input signal. This approach results in the so-called *generalized* memory polynomial that relates the output y_{GMP} to its input u by

$$\begin{aligned}
 y_{\text{GMP}}(m) = & \sum_{n=1}^{N_a} \sum_{l=0}^{M_a} a_{nl} u(m-l) |u(m-l)|^{n-1} \\
 & + \sum_{n=2}^{N_b} \sum_{l=0}^{M_b} \sum_{v=1}^V b_{nlv} u(m-l) |u(m-l-v)|^{n-1} \\
 & + \sum_{n=2}^{N_c} \sum_{l=0}^{M_c} \sum_{w=1}^W c_{nlw} u(m-l) |u(m-l+w)|^{n-1}
 \end{aligned} \tag{3.72}$$

The output of the generalized memory polynomial model is composed of three parts. The first part corresponds to the original memory polynomial model and uses time-aligned input samples. The second and third part introduces cross-terms between the input signal and its lagging or leading envelope terms up to the V^{th} and W^{th} order, respectively. Typically, the nonlinearity orders and the memory depths of the leading and lagging cross-terms (N_b, N_c, M_b , and M_c) are significantly smaller than N_a and M_a of the time-aligned part. Furthermore, the orders of the lagging and leading envelope signal shifts (V and W) are commonly relatively low. As the model is also linear in the parameters, an extended form of (3.71) can be used to estimate all the coefficients a_{nl} , b_{nlv} , and c_{nlw} .

By considering these cross-terms, the generalized memory polynomial is positioned between the memory polynomial model and the original Volterra series in terms of model complexity and performance. The generalized memory polynomial is seen as a practical alternative to the computationally heavy Volterra series and is attractive when the system exhibits strongly nonlinear memory effects. If this is the case, the generalized memory polynomial outperforms its traditional time-aligned counterpart and the added model complexity is justified. However, when no strong nonlinear effects are present, both models perform in a comparable way.

3.5 Back to the case study

Now that you have digested the mathematical framework underlying the Volterra theory, we go back to the case study and apply the nonlinear modelling to the power amplifier measurements. The aim of this section is to model the spectral regrowth that is visible in the bottom row of Figure 1.8. As the power amplifier operates at high frequencies, we can only measure the linear behaviour and the in-band odd-order nonlinear distortion around the carrier frequency.

Due to the high number of 66 excited frequencies, it is not possible to identify all the coefficients $h_n(l_1, \dots, l_n)$ of the original Volterra series formulation in (3.63). A quick back-of-the-envelope calculation reveals that the 5th-order Volterra kernel alone requires the estimation of $66^5 \approx 1.25 \times 10^9$ contributions. And while this number can be greatly reduced if we consider symmetric kernels, it still remains an astonishing large amount of coefficients. Don't forget that we need to acquire at least the same amount of data as the number of coefficients to be able to estimate all of them.

To mitigate this combinatorial coefficient explosion, we will fall back to the memory polynomial model introduced in the previous section. With the measurement setup we have gathered a total of $N = 4096$ time-domain samples of $u(m)$ and $y(m)$. Afterwards, these input-output time-domain samples have been used to estimate a memory polynomial model with parameters $N_{\text{order}} = 5$ and $M_{\text{lag}} = 9$. This parameter choice results in a total of 50 parameters that need to be estimated. The results of this estimation are shown in Figure 3.6. From the time-domain results, one can see that the model results in a fairly accurate prediction (RMS error of 8.67%) with error bounds that remain constant along the time axis. However, comparing the model with the measurement results in the frequency domain reveals a different picture. The model is capable of accurately predicting the output at the excited frequencies and in a bounded frequency region centered around the excitation bandwidth. As these tones contain most of the energy, it is a positive sign that the model can capture the nonlinear effects at those frequencies. However, at tones further away from the excitation bandwidth, the model loses its accuracy. There are two main reasons for this reduced modelling accuracy. First of all, the memory polynomials are an approximation of the original Volterra series and therefore do not inherit the same modelling capabilities. Secondly, the power amplifier contains feedback mechanisms that

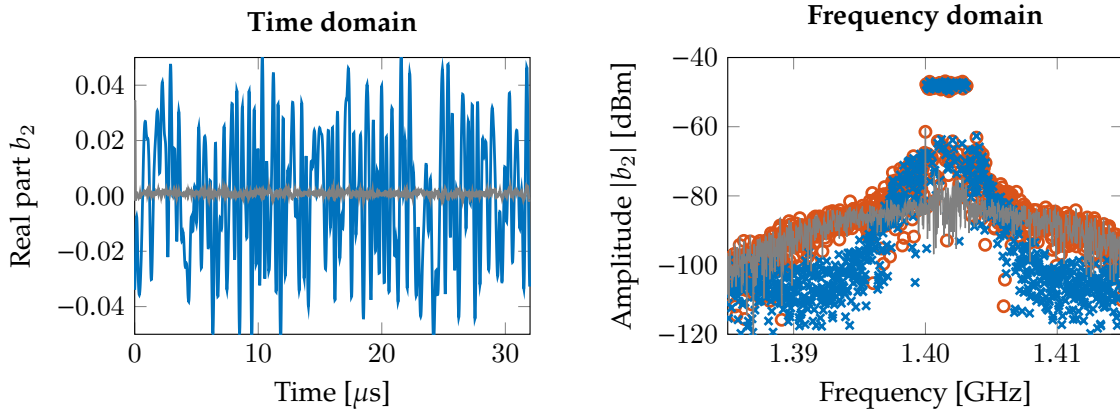


Figure 3.6: Modelling results of the output wave b_2 . In blue: the measurement data, in red: the modelled output, and in gray: the error.

are not included in the Volterra theory.

3.6 Relating the Best Linear Approximation to the Volterra theory

In Chapter 2, we explored the BLA theory and how it derives an approximate linear model of a nonlinear system under modulated excitation. This section links the Volterra theory to the BLA framework and provides you with more intuition and understanding on how the nonlinear distortion impacts the BLA. To do so, we first need to make a distinction between coherent and non-coherent contributions in the output spectrum of a nonlinear system.

As the name suggest, coherent contributions are those nonlinear contributions that maintain coherency between the input and the resulting output of the nonlinear system. Mathematically, coherency at a single frequency is defined as having a constant phase relationship between the input and output

$$\angle Y(\omega) = \angle U(\omega) + \varphi_0 \quad (3.73)$$

where φ_0 is a fixed phase offset inherent to the behaviour of the system. All output contributions that do not satisfy (3.73), are simply called non-coherent contributions. For a linear system $G(j\omega)$ that is excited by a steady-state periodic excitation, all output contributions are coherent, irrespective of the phase realisation of the input $U(\omega)$. φ_0 in

that case is nothing else than $\angle G(j\omega)$. For a nonlinear system, this is no longer generally true due to the frequency recombination mechanism. To better illustrate the difference between coherent and non-coherent contributions generated by a nonlinear system, let us examine the behaviour of a third-order and second-order Volterra kernel in more detail.

In Section 3.3.3, we derived that the third-order Volterra kernel results in the following output contributions in the frequency domain

$$Y_3(\omega) = \frac{1}{(2\pi)^2} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} H_3(j\omega_1, j\omega_2, j\omega - j\omega_1 - j\omega_2) U(\omega_1) U(\omega_2) U(\omega - \omega_1 - \omega_2) d\omega_1 d\omega_2 \quad (3.74)$$

The question we now need to answer is which of these contributions are (non-)coherent? From (3.73), we find that a certain frequency combination is coherent if and only if

$$\angle U(\omega_1) U(\omega_2) U(\omega - \omega_1 - \omega_2) = \angle U(\omega) \quad (3.75)$$

This coherency condition is only satisfied if $\omega_1 = -\omega_2$. For a complex-conjugate input spectrum, we then obtain

$$\angle U(\omega_1) U(-\omega_1) U(\omega) = |U(\omega_1)|^2 \angle U(\omega) \quad (3.76)$$

All other frequency combinations where $\omega_1 \neq -\omega_2$ result in a non-coherent output contribution.

Taking a look at the expression of the output contributions for a second-order Volterra kernel

$$Y_2(\omega) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} H_2(j\omega_1, j\omega - j\omega_1) U(\omega_1) U(\omega - \omega_1) d\omega_1 \quad (3.77)$$

reveals that a second-order kernel expresses coherent contributions for $\omega_1 = 0$. So a DC excitation $U(0)$ is required to obtain coherency. Unfortunately, a DC excitation modifies the nonlinear regime of the system as it imposes a different bias operating point. This change in operating point will fundamentally change the nonlinear behaviour of the system and the associated Volterra series approximation. Therefore, in most applications it is considered that $U(0) = 0$. As a consequence, the second-order Volterra kernel only results in non-coherent output contributions. In general, odd-order nonlinearities produce both coherent and non-coherent output contributions. If DC is not excited, even-order nonlinearities always generate non-coherent contributions.

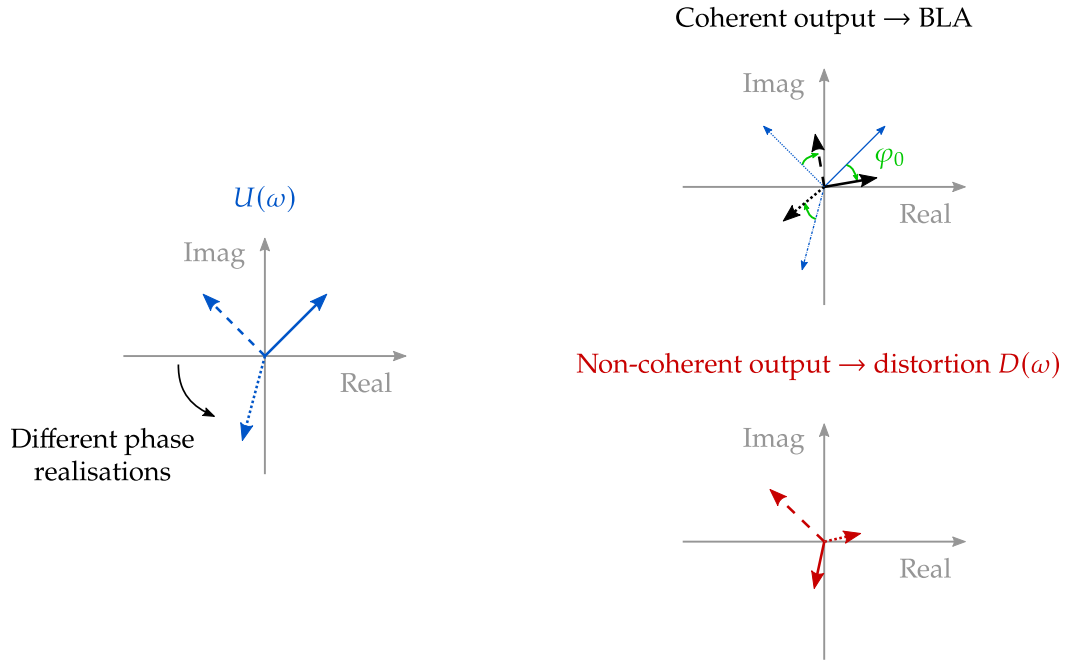


Figure 3.7: The **BLA** captures the coherent input-output behaviour, while the non-coherent part of the output behaves as a 'random' source that depends on the input phase realisation.

But why do we need to consider coherency and why is it important for the **BLA** theory? Well, essentially the **BLA** models the behaviour of the input to the coherent part of the output. This insight is best explained by revisiting the earlier introduced *robust method* (Section 2.4). Recall that the idea of the robust method is to combine several phase realisations of a multisine excitation with stochastic averaging to retrieve the **BLA** and the variance of the nonlinear distortion. This method heavily relies on the coherence property to function properly (Figure 3.7). As the coherent part of the output 'follows' the input phase, the coherent input-output behaviour remains constant across the different phase realisations of the input. Therefore, averaging relative to the input phase realisation is possible and results in the **BLA**. The non-coherent part of the output, represented by the non-linear distortion term $D(\omega)$, does change with the input phase realisation and introduces a 'stochastic' component to the output. Due to its 'stochastic' nature, it makes sense to compute the variance across the different phase realisations to obtain a quantitative measure about As a result, taking the variance across the different phase realisations provides a quantitative measure about the significance of the nonlinear distortion.

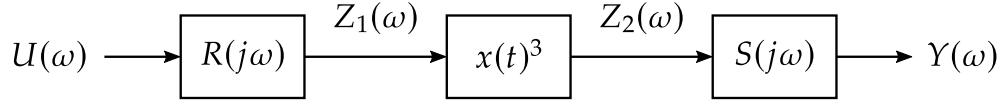


Figure 3.8: Wiener-Hammerstein system that consists of a cubic nonlinearity sandwiched between two *LTI* systems $R(j\omega)$ and $S(j\omega)$.

Example

To further increase understanding, let us consider the Wiener-Hammerstein example in Figure 3.8 and calculate the *BLA* starting from the coherent output contributions. In a first stage, we need to compute the multidimensional Volterra relationship between the input $U(\omega)$ and the output $Y(\omega)$. For the Wiener-Hammerstein system we obtain the following signal relationships

$$\begin{aligned} Z_1(\omega) &= R(j\omega) U(\omega) \\ Z_2(\omega_1 + \omega_2 + \omega_3) &= R(j\omega_1) R(j\omega_2) R(j\omega_3) U(\omega_1) U(\omega_2) U(\omega_3) \\ Y(\omega_1 + \omega_2 + \omega_3) &= S(j\omega_1 + j\omega_2 + j\omega_3) R(j\omega_1) R(j\omega_2) R(j\omega_3) U(\omega_1) U(\omega_2) U(\omega_3) \end{aligned} \quad (3.78)$$

where $\omega = \omega_1 + \omega_2 + \omega_3$. In the previous section, we have already derived that for a third-order kernel we need to take $\omega_2 = -\omega_1$ to obtain the coherent output contributions. Applied to this example, we then obtain for the coherent part of the output

$$\begin{aligned} Y_{\text{coherent}}(\omega) &= \frac{1}{(2\pi)^2} \int_{-\infty}^{+\infty} S(j\omega) R(j\omega_1) R(-j\omega_1) R(j\omega) U(\omega_1) U(-\omega_1) U(\omega) d\omega_1 \\ &= \frac{1}{(2\pi)^2} R(j\omega) S(j\omega) \left(\int_{-\infty}^{+\infty} |R(j\omega_1)|^2 |U(\omega_1)|^2 d\omega_1 \right) U(\omega) \end{aligned} \quad (3.79)$$

The *BLA* of the system is retrieved by dividing the the coherent part of the output spectrum by the input spectrum, which results in

$$G^{\text{BLA}}(j\omega) = \frac{1}{(2\pi)^2} \underbrace{R(j\omega) S(j\omega)}_{\text{linear dynamics}} \underbrace{\left(\int_{-\infty}^{+\infty} |R(j\omega_1)|^2 |U(\omega_1)|^2 d\omega_1 \right)}_{\text{real constant}} \quad (3.80)$$

The resulting *BLA* consists of the product of two terms. The first term represents the underlying linear dynamic behaviour, and the second term is a real constant that introduces

the signal dependency in the [BLA](#). Remark that the [BLA](#) is independent of the phase realisation of the input, as it should be.

3.7 Summary

This chapter introduced you to the wonderful (and sometimes quite mathematical) world of Volterra. We have seen that the Volterra theory elegantly generalizes linear systems theory to enable the analysis of nonlinear dynamic systems. At its core, the theory uses a functional series approximation to include nonlinear effects in the modelling of the system's behaviour. By doing so, notions such as causality and stability could be straightforwardly extended towards nonlinear systems.

The Volterra model can be represented both in the time domain and frequency domain. While for linear dynamic systems the transition towards the frequency domain makes the analysis easier (convolution becomes a product), this simplification is only valid in the multidimensional Fourier space. When transitioning towards the actual output spectrum, multi-dimensional convolutions are still involved. These convolutions are the main cause of the frequency recombination mechanism that is so inherent to nonlinear systems. In spite of its advantages, the Volterra model is plagued by a combinatorial explosion of model coefficients, which makes the estimation almost impossible. To resolve this issue, we have studied several block-structured models that approximate the original Volterra series. They trade model complexity with ease of estimation and are widely used in practice to obtain a nonlinear model with a limited number of model coefficients. In the next chapter, we will explore how these nonlinear models can be used to solve application-specific problems.

Chapter 4

Applications of Volterra theory

In the last two chapters, we have revealed how to describe the behaviour of a nonlinear system with an approximating model. We have covered both a linearized (BLA) and full-blown nonlinear (Volterra) modelling framework. Unfortunately, a model on its own is not very interesting. From an engineering perspective, a model only becomes valuable if we can use it to gain physical insight or solve engineering problems. This chapter delves into the discussion and analysis of one application, aiming to further illustrate the practical applicability of the Volterra series.

4.1 Nonlinear predistortion

As established in earlier chapters, many systems demonstrate nonlinear behaviour that can be described by a Volterra series. An interesting aspect to study is whether it is possible to linearize the nonlinear dynamic nature of the Volterra series. This is especially important for telecommunication systems, where transmit power amplifiers are supposed to have an amplification that is as linear as possible. A nonlinear power amplifier gives rise to spectral regrowth outside the original input excitation band, thereby leading to interference in adjacent radio channels. Predistortion is a technique that mitigates the nonlinear distortion, which ultimately results in a more linear system overall. To reach this objective, a predistorter is added in front of the nonlinear system (Figure 4.1). The

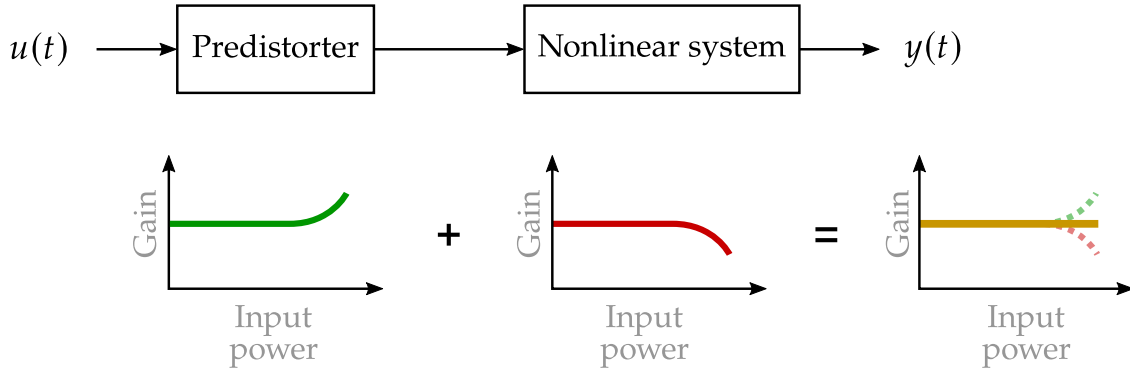


Figure 4.1: Predistortion modifies the input signal of the nonlinear system to counteract the nonlinear characteristic and linearize the overall system response.

predistorter introduces "inverse distortion" to the initial input signal, effectively counteracting the nonlinear behavior of the system. This technique has become standard practice in telecommunication systems, as it allows for the extraction of additional usable power from amplifiers. This eliminates the need for constructing larger, less efficient, and more costly amplifiers.

Although the fundamental concept of predistortion appears straightforward, a critical question that remains unanswered is how to derive and implement this predistorter. To be effective, the predistorter must possess the capability to accurately inverse the behavior of the nonlinear system. Regrettably, not all nonlinear systems have an inverse. Furthermore, among those nonlinear systems that do possess an inverse, it is often only defined within a restricted range of input amplitudes. However, it will become apparent that even in such cases, a p^{th} -order inverse exists without any limitation on the input amplitude range. The p^{th} -order inverse of a nonlinear Volterra system $\mathcal{H}$ is characterized as a system that, when connected in series with $\mathcal{H}$, yields a system where the second through p^{th} -order Volterra kernels are exactly zero. If we denote $\mathcal{Q}$ as the Volterra series representation of the p^{th} -order inverse connected in series with the nonlinear system $\mathcal{H}$, then the Volterra series of $\mathcal{Q}$ can be expressed as follows:

$$\mathcal{Q}[u(t)] = u(t) + \sum_{n=p+1}^{+\infty} \mathcal{Q}_n[u(t)] \quad (4.1)$$

where $\mathcal{Q}_n$ is the n^{th} -order Volterra operator of the system $\mathcal{Q}$, as defined in (3.15).

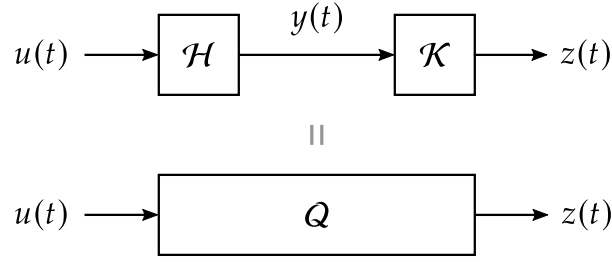


Figure 4.2: Series connection of two Volterra systems.

In this section, we discuss the necessary and sufficient conditions for the existence of an inverse of a Volterra system, as well as how to obtain such an inverse. We will first discuss how to obtain the p^{th} -order post-inverse. Subsequently, we will demonstrate that the p^{th} -order post-inverse and p^{th} pre-inverse are equivalent. This establishes the foundation for a concrete definition of the p^{th} -order inverse that can then be used for predistortion purposes.

4.1.1 Series connection of nonlinear systems

Before we can formally introduce the p^{th} -order inverse, we first need to study how the series connection of two nonlinear systems behaves. To do so, we will analyze the relationship between the global Volterra series of a series connection Q and the Volterra series $\mathcal{H}$ and $\mathcal{K}$ of the individual components within this connection, as shown in Fig. 4.2. We explicitly assume that there is no interaction between the two systems, meaning there are no loading effects from one system onto the other. The relationship between the input $u(t)$ and the response $y(t)$ of the Volterra system $\mathcal{H}$ in Fig. 4.2 is given by

$$y(t) = \sum_{n=1}^{+\infty} \mathbb{H}_n[u(t)] \quad (4.2)$$

The output $z(t)$ resulting from the series connection of systems $\mathcal{H}$ and $\mathcal{K}$ can subsequently be calculated as follows

$$z(t) = \sum_{m=1}^{+\infty} \mathbb{K}_m[y(t)] = \sum_{m=1}^{+\infty} \mathbb{K}_m \left[\sum_{n=1}^{+\infty} \mathbb{H}_n[u(t)] \right] \quad (4.3)$$

Additionally, the output $z(t)$ can also be characterized by the Volterra series of the system Q in Fig. 4.2

$$z(t) = \sum_{r=1}^{+\infty} Q_r[u(t)] \quad (4.4)$$

To determine the operators Q_r in terms of the operators H_n and K_m , we will use the power series representation of the Volterra series that is obtained if we scale the input with the scaling factor A . Performing the substitution $u(t) \rightarrow A u(t)$ in (4.4) results in

$$z(t) = \sum_{r=1}^{+\infty} Q_r[A u(t)] = \sum_{r=1}^{+\infty} A^r Q_r[u(t)] \quad (4.5)$$

The above expression is a power series in A with the coefficient of A^r being $Q_r[u(t)]$. A similar power series is obtained if we replace $u(t)$ by $A u(t)$ in (4.3)

$$\begin{aligned} z(t) &= \sum_{m=1}^{+\infty} K_m[y(t)] = \sum_{m=1}^{+\infty} K_m \left[\sum_{n=1}^{+\infty} H_n[A u(t)] \right] \\ &= \sum_{m=1}^{+\infty} K_m \left[\sum_{n=1}^{+\infty} A^n H_n[u(t)] \right] \\ &= \sum_{m=1}^{+\infty} K_m \left[\sum_{n=1}^{+\infty} A^n y_n(t) \right] \\ &= \sum_{m=1}^{+\infty} \sum_{n_1=1}^{+\infty} \cdots \sum_{n_m=1}^{+\infty} A^{n_1+\cdots+n_m} K_m\{y_{n_1}, \dots, y_{n_m}\} \end{aligned} \quad (4.6)$$

The last step in the above derivation makes use of the integral representation of K_m in which

$$K_m\{y_1, \dots, y_{n_m}\} = \int_{-\infty}^{+\infty} k_m(\tau_1, \dots, \tau_m) y_{n_1}(t - \tau_1) \cdots y_{n_m}(t - \tau_m) d\tau_1 \cdots d\tau_m \quad (4.7)$$

From the expressions (4.5) and (4.6), we can now obtain the Volterra operators Q_r in terms of H_n and K_m by equating the coefficients belonging to the same powers of A . To do so, we first observe in (4.6) that the minimum value of the sum $n_1 + n_2 + \cdots + n_m$ is m . The first power in (4.3) thus can only originate from the term with $m = 1$ and $n_1 = 1$. Equating the coefficients belonging to A^1 in (4.5) and (4.6) results in the first relationship

$$Q_1[u] = K_1[y_1] = K_1 H_1[u] \quad (4.8)$$

Based on the expression above, we can deduce that the linear component in the series connection of two Volterra systems is essentially the successive application of the linear operators within these systems (Figure 4.3a). Equating the coefficients in A^2 results in two contributions: one for $m = 1$ with $n_1 = 2$ and one for $m = 2$ with $n_1 = n_2 = 1$. This result is in the following expression for the second-order operator

$$\mathbb{Q}_2[u] = \mathbb{K}_1[y_2] + \mathbb{K}_2[y_1] = \mathbb{K}_1\mathbb{H}_2[u] + \mathbb{K}_2\mathbb{H}_1[u] \quad (4.9)$$

The resulting block diagram of the operator $\mathbb{Q}_2$ consists of two branches, as visualized in Figure 4.3b.

The calculation of the operator $\mathbb{Q}_3$ involves one additional aspect that is not encountered with the determination of $\mathbb{Q}_1$ and $\mathbb{Q}_2$. Similar as before, we first equate the coefficients belonging to A^3 . Remark that in (4.6) the third-power contributions are obtained for $m = 1$ with $n_1 = 3$, for $m = 2$ with $n_1 = 1$ and $n_2 = 2$ and also with $n_1 = 2$ and $n_2 = 1$, and finally for $m = 3$ with $n_1 = n_2 = n_3 = 1$. These contributions result in the following third-order Volterra operator

$$\mathbb{Q}_3[u] = \mathbb{K}_1[y_3] + \mathbb{K}_2\{y_1, y_2\} + \mathbb{K}_2\{y_2, y_1\} + \mathbb{K}_3[y_1] \quad (4.10)$$

Since a Volterra operator can be considered symmetric without loss of generality, we shall assume that the operators $\mathbb{K}_n$ are symmetric such that

$$\mathbb{K}_2\{y_1, y_2\} = \mathbb{K}_2\{y_2, y_1\} \quad (4.11)$$

Taking into account symmetry enables us to rewrite (4.10) to

$$\mathbb{Q}_3[u] = \mathbb{K}_1[y_3] + 2\mathbb{K}_2\{y_1, y_2\} + \mathbb{K}_3[y_1] \quad (4.12)$$

In the expression above, the terms $\mathbb{K}_1[y_3]$ and $\mathbb{K}_3[y_1]$ can be synthesized by consecutive application of the corresponding $\mathbb{H}_n$ and $\mathbb{K}_m$ Volterra operators. However, the determination of $\mathbb{K}_2\{y_1, y_2\}$ is not straightforward. This is the additional aspect mentioned previously. Making use of the bilinear properties of the $\mathbb{K}_2$ operator, the following identity can be used to determine $\mathbb{K}_2\{y_1, y_2\}$

$$\mathbb{K}_2[y_1 + y_2] = \mathbb{K}_2[y_1] + 2\mathbb{K}_2\{y_1, y_2\} + \mathbb{K}_2[y_2] \quad (4.13)$$

Substituting (4.13) in (4.12) results in the following final expression for $\mathbb{Q}_3$ in operator notation (Figure 4.3c)

$$\mathbb{Q}_3 = \mathbb{K}_1\mathbb{H}_3 + \mathbb{K}_2[\mathbb{H}_1 + \mathbb{H}_2] - \mathbb{K}_2\mathbb{H}_1 - \mathbb{K}_2\mathbb{H}_2 + \mathbb{K}_3\mathbb{H}_1 \quad (4.14)$$

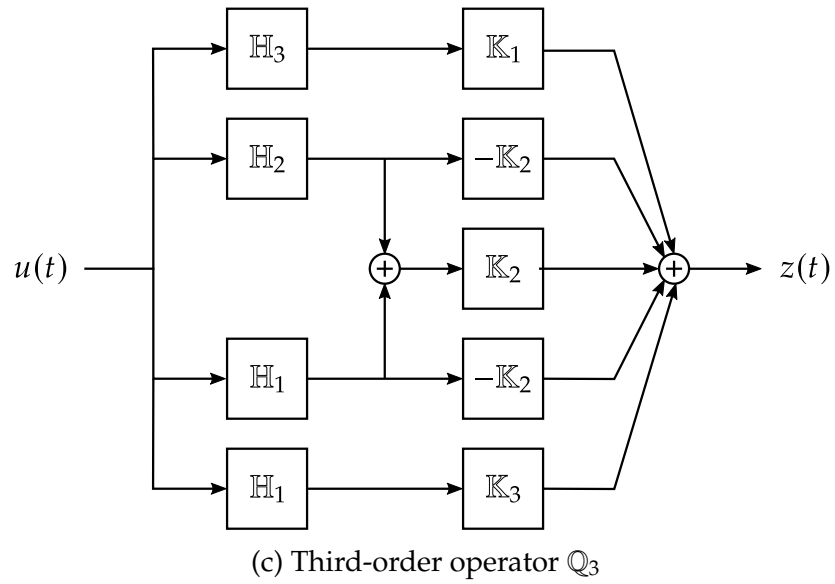
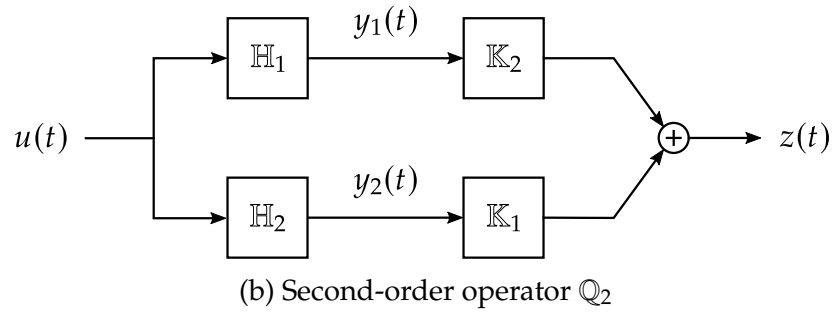
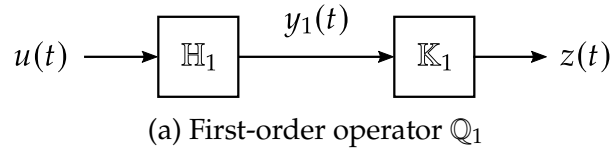


Figure 4.3: Block diagrams of the operators $\mathbb{Q}_1$, $\mathbb{Q}_2$, and $\mathbb{Q}_3$.

Note that, due to the multilinear nature of the Volterra operators, the operator notation does not permit argument splitting. For instance, $\mathbb{K}_2[\mathbb{H}_1 + \mathbb{H}_2]$ is not equal to $\mathbb{K}_2[\mathbb{H}_1] + \mathbb{K}_2[\mathbb{H}_2]$. The procedure we outlined here for the determination of $\mathbb{Q}_1$, $\mathbb{Q}_2$, and $\mathbb{Q}_3$, can be used to determine each of the operators $\mathbb{Q}_n$.

4.1.2 The p^{th} -order post-inverse

Having now established how the Volterra series of a series connection can be expressed in terms of the Volterra series of the connecting systems, we can employ this insight to compute the p^{th} -order inverse of the system $\mathcal{H}$, denoted as $\mathcal{H}_{(p)}^{-1}$. This is achieved by identifying the Volterra operators $\mathbb{K}_n$ in such a way that the series connected system $\mathcal{Q}$ satisfies the conditions specified by (4.1).

The first condition to be satisfied in (4.1) refers to the linear behavior and boils down to

$$\mathbb{Q}_1[u(t)] = \mathbb{K}_1[\mathbb{H}_1[u(t)]] = u(t) \quad (4.15)$$

It is clear that this condition is satisfied by a linear operator $\mathbb{K}_1$ which is the inverse of the causal linear operator $\mathbb{H}_1$. In the frequency domain, taking the inverse is straightforward and the requirement becomes

$$K_1(j\omega) = \frac{1}{H_1(j\omega)} \quad (4.16)$$

where K_1 and H_1 are the Fourier transforms of the Volterra kernels $k_1(\tau)$ and $h_1(\tau)$, respectively. Note that it is not always possible to satisfy (4.16) with a stable and causal linear operator K_1 . Under the assumption that H_1 is a rational function of $j\omega$, stable causal behavior of K_1 can only be obtained if all the zeros of H_1 lie in the left-half plane. We assume for the remainder of this section that the linear operator that satisfies (4.16) is stable and causal.

The second condition we need to satisfy from (4.1) is that $\mathbb{Q}_2[u] = 0$. Using the operator notation in (4.9), the condition becomes

$$\mathbb{Q}_2[u] = \mathbb{K}_1\mathbb{H}_2[u] + \mathbb{K}_2\mathbb{H}_1[u] = 0 \quad (4.17)$$

This condition will only be satisfied for all u if the operator $\mathbb{K}_2$ satisfies

$$\mathbb{K}_2\mathbb{H}_1 = -\mathbb{K}_1\mathbb{H}_2 \quad (4.18)$$

Solving this equation for $\mathbb{K}_2$ requires post-multiplication of both sides by the inverse of the operator $\mathbb{H}_1$

$$\begin{aligned}\mathbb{K}_2 &= -\mathbb{K}_1\mathbb{H}_2\mathbb{H}_1^{-1} \\ &= -\mathbb{K}_1\mathbb{H}_2\mathbb{K}_1\end{aligned}\tag{4.19}$$

Again, the operator $\mathbb{K}_2$ is causal and stable if the operators $\mathbb{K}_1$ and $\mathbb{H}_2$ are causal and stable.

We follow a similar approach to determine each of the operators up to order p for the system $\mathcal{H}_{(p)}^{-1}$. For example, the operator $\mathbb{K}_3$ that satisfies $\mathbb{Q}_3[u] = 0$ can be computed as follows

$$\begin{aligned}\mathbb{K}_3 &= -\mathbb{K}_1\mathbb{H}_3\mathbb{H}_1^{-1} - \mathbb{K}_2[\mathbb{H}_1 + \mathbb{H}_2]\mathbb{H}_1^{-1} + \mathbb{K}_2\mathbb{H}_1\mathbb{H}_1^{-1} + \mathbb{K}_2\mathbb{H}_2\mathbb{H}_1^{-1} \\ &= -\mathbb{K}_1\mathbb{H}_3\mathbb{K}_1 - \mathbb{K}_2[\mathbb{H}_1 + \mathbb{H}_2]\mathbb{K}_1 + \mathbb{K}_2\mathbb{H}_1\mathbb{K}_1 + \mathbb{K}_2\mathbb{H}_2\mathbb{K}_1\end{aligned}\tag{4.20}$$

It is always possible for an operator $\mathbb{K}_n$ to be isolated from the condition $\mathbb{Q}_n = 0$. This is because the only term in the expression for $\mathbb{Q}_n[u]$ which involves the operator $\mathbb{K}_n$ is $\mathbb{K}_n\mathbb{H}_1[u]$. All the other terms in the expression require $\mathbb{K}_j$, with $j < n$, which all have already been determined. Furthermore, the operators $\mathbb{K}_n$ are all causal and stable since the system $\mathcal{H}$ and the operator $\mathbb{K}_1$ is causal and stable. Thus, to obtain a causal and stable p^{th} -order post-inverse $\mathcal{H}_{(p)}^{-1}$ it is sufficient that the inverse of the linear operator $\mathbb{H}_1$ is stable and causal.

4.1.3 The p^{th} -order pre-inverse

So far, our focus has been on determining the p^{th} -order post-inverse and understanding how it can be calculated using the Volterra operators $\mathbb{H}_n$ of the nonlinear system $\mathcal{H}$. However, as noted in the introduction of this section, our objective is to have the capability to predistort the nonlinear system. This necessitates the computation of a pre-inverse for the nonlinear system, rather than a post-inverse. We know that the series connection of two nonlinear systems generally affects the overall system response. As a result, connecting the pre- or post-inverse in series with $\mathcal{H}$ does not result in the same system (Figure 4.4). Nevertheless, it can be shown (without proof) that when the pre-inverse and post-inverse of order p are connected in series with the nonlinear system $\mathcal{H}$, the Volterra operators of the resulting cascaded systems are the same up to the p^{th} order. This implies that the p^{th} -order pre-inverse and post-inverse of the system are equivalent up to the p^{th} -order Volterra

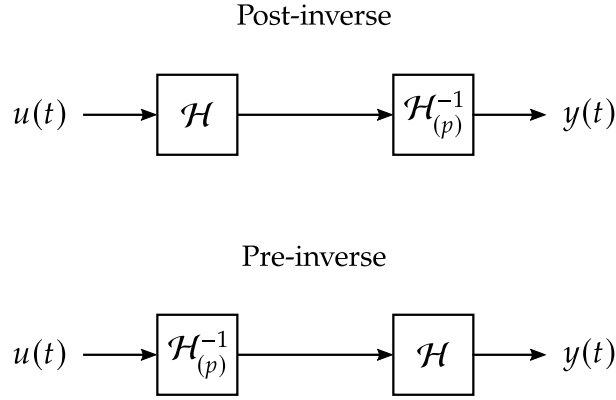


Figure 4.4: Post- and pre-inverse configurations of the system $\mathcal{H}$.

kernel. We can therefore use the p^{th} -order post-inverse as a predistorter. However, this equivalence does not mean that for orders larger than p the Volterra operators of these cascaded systems are the same. Only if we take the limit $p \rightarrow \infty$ of $\mathcal{H}_{(p)}^{-1}$ can we truly refer to the inverse $\mathcal{H}^{-1}$ of the system $\mathcal{H}$.

4.1.4 Digital predistortion of a power amplifier

One application of the p^{th} -order inverse is the linearization of a given nonlinear system. As mentioned earlier in the introduction, this linearization is especially important within telecommunication systems where it is key to linearize the behaviour of the power amplifier. The aim of this subsection is to evaluate the performance of the so-called **Digital PreDistortion (DPD)** technique applied to a simulation example of an **RF** power amplifier (Figure 4.5). The **Device Under Test (DUT)** is an RF power amplifier operating around 2 GHz and has been designed in an integrated GaN-technology. The reference signal a_{ref} is a random-phase multisine centred around 2 GHz with a 100 MHz bandwidth.

The **DPD** technique implements the nonlinear predistortion in the digital domain by predistorting the input signal a_{ref} before feeding it to the amplifier. The estimation of the post-inverse of the amplifier requires to sense the input wave a_1 and the output wave b_2 by means of couplers. The resulting measured waves a_1^m and b_2^m are then used to estimate the post-inverse, which is obtained by inverse modelling from the output wave b_2 to the input wave a_1 . The accuracy of the post-inverse estimation is directly proportional to the model type that is used to capture the nonlinear effects. Following the same rationale as

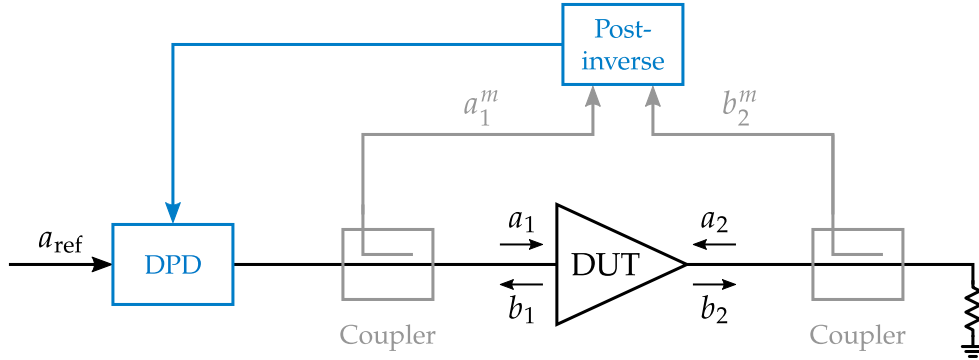


Figure 4.5: Digital predistortion architecture that aims at predistorting the nonlinear behaviour of the *DUT*.

discussed in Section 3.4, we can use block-structured models to obtain a nonlinear model with a limited number of parameters that still provides high accuracy. For the post-inverse estimation we use the generalized memory polynomials, which were introduced in Section 3.4.4, with the following settings: $N_a = 5$, $N_b = N_c = 2$, and $M_a = M_b = M_c = V = W = 1$. To obtain an accurate digital predistortion, the post-inverse estimate needs to be iteratively updated. The reason for this is that the nonlinear behavior of the power amplifier is signal dependent. Once the *DPD* is applied, the original input signal changes and in turn alters the nonlinear response of the amplifier. Consequently, multiple iterations are necessary to ensure that the post-inverse model accurately accounts for all nonlinear effects. Figure 4.5 shows the output spectrum of the power amplifier, comparing the scenarios with and without *DPD*. It is clear that using *DPD* eliminates the spectral regrowth, leading to a significantly more linear response. Thanks to the straightforward implementation of these *DPD* algorithms in digital signal processing hardware, they find application in nearly every modern wireless telecommunication system, enhancing linearity requirements.

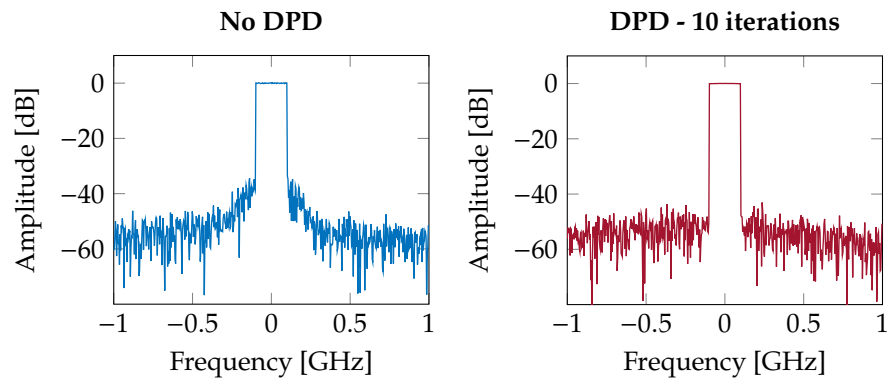


Figure 4.6: Application of *DPD* effectively compensates for the nonlinear behaviour of the *DUT*.